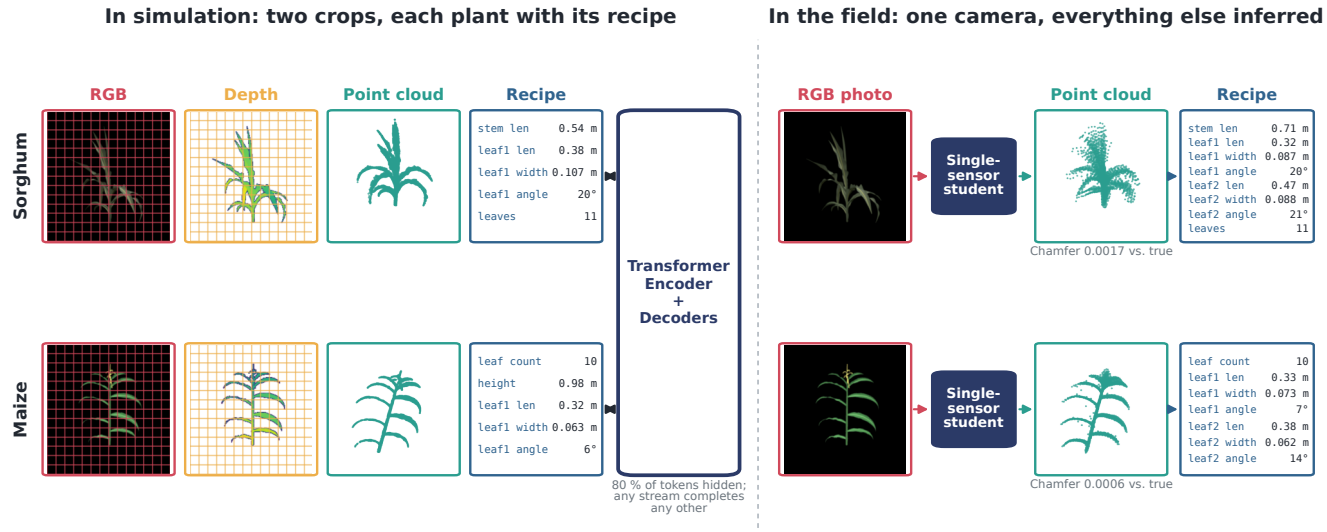
# Procedural Reconstruction of Slender Plants from Any Field Modality

Anonymous CVPR submission

Paper ID *****




*_1_plant_0004_04 (raw files in paper/data); right: real outputs, sorghum zero-shot (pretrained arm), maize distilled; no ground truth shown. RGB shown as the current renders (brightened sorghum, background-free maize); each m*

Figure 1. **Any sensor in, procedural plant out.** From any subset of field-practical modalities (a single photograph, a depth frame, or a point cloud truncated by row neighbors), CROPCODE recovers the full procedural representation of the plant: a kilobyte-scale, physiologically interpretable parameter set that can be re-rendered to complete geometry, used as simulation input, and used for trait analytics at field scale.

## Abstract

*Editable, biologically interpretable 3D models of field-grown crops support high-throughput phenotyping,* in silico *breeding trials, and agronomic digital twins, but recovering such models from field sensor data remains difficult: plant organs are slender, neighboring plants occlude one another, and a field campaign rarely yields more than a single photograph, a noisy depth frame, or a partial point cloud per plant. Slender geometry under occlusion is a regime in which learned 3D completion and Chamfer-based evaluation both fail. In this work, we present* CROPCODE*, a multimodal masked-autoencoder framework that recovers the complete procedural representation of a plant, the generator parameters encoding leaf count, insertion heights, blade angles, and surface geometry, from any subset of RGB, depth, and point-cloud input in a single forward pass. The model is pretrained on 15,000 procedurally generated plants per species (maize and sorghum), fine-tuned on rows in which simulated neighbors occlude the target, and distilled from a full-modality teacher into single-sensor students. Distillation lowers the cross-modal generation loss by 30% in sorghum and 44% in maize, and halves the sorghum parameter error from a point cloud alone. Training on rows removes 92% (maize) and 85% (sorghum) of the error that neighbors add for a model trained on isolated plants. Controlled ablations in both species show that exposing most of the procedural parameters to the encoder degrades phenotype recovery, while using them only as a reconstruction target degrades the sorghum representation and improves the maize one. We further show that Chamfer distance and linear probes on the frozen representation rank input modalities differently on slender geometry, and we adopt the probe as the deciding metric. On real field scans with independently reconstructed ground truth,* CROPCODE *matches the accuracy of a staged annotation pipeline and reduces the time required for procedural reconstruction from hours of manual specification, or*

*minutes of pipeline computation, to milliseconds per plant. We will release* FIELDPROC*, the 30,000-plant, two-species benchmark behind these experiments, which pairs multimodal field captures under simulated row occlusion with exact procedural ground truth.*

## 1. Introduction

Accurate three-dimensional models of individual crop plants enable high-throughput phenotyping of morphological traits such as leaf angle, internode length, and canopy architecture, and they are the basis of emerging agricultural digital twins in which breeding trials run *in silico* [19, 25]. The representation best suited to these uses is procedural: a compact set of biologically interpretable parameters that a generator expands into complete, editable, simulation-ready geometry [20, 36]. A procedural plant occupies a few kilobytes where a laser scan occupies megabytes, every parameter corresponds to a trait used by agronomists, and the model can be edited, re-rendered, and perturbed in ways that unstructured geometry does not support. Field crops also instantiate a challenge of general interest in 3D computer vision: recovering a parametric model of a thin, heavily occluded object from partial multimodal observations.

Yet recovering procedural structure from field-acquired sensor data remains a significant challenge, and every existing route presupposes a rich, clean observation. Manual parameter specification takes hours per plant [20]. Staged reconstruction pipelines automate the process; a recent system annotates laser scans with a vision-language model and fits a procedural descriptor to 5.4 mm median accuracy, but it requires a complete scan of an isolated plant and roughly ten minutes of computation per plant [18]. Per-scene optimization methods search generator parameters against multi-view imagery [53], and recent learned models infer parametric plants, but only from a clean, complete point cloud [8, 14]. A phenotyping campaign produces the opposite: thousands of plants per day, each observed by whichever sensor was practical (a camera, a depth frame, or a partial scan).

In addition, crop plants are slender: a maize blade is centimeters wide, millimeters thick, and up to a meter long. In a planted row, each plant is further occluded by its neighbors, whose leaves interleave with the target; in our field simulations, a third of the target's pixels and two thirds of the scene's point cloud belong to neighboring plants. This regime violates the assumptions of learned 3D completion: the standard benchmarks contain compact, largely convex objects and no slender-under-occlusion category [50, 52], and Chamfer distance, the default metric, is structurally insensitive to thin sheets, so it can reward reconstructions that miss entire blades [43, 48]. Occlusion-aware completion of thin plant organs has begun to address the input side of this regime, but its output is a completed point cloud rather than structure [27]. To our knowledge, no prior method recovers a procedural, re-renderable representation of slender crop organs from a single field-practical observation (a photograph, a depth frame, or a partial point cloud) under row-neighbor occlusion: methods that infer plant parameters require clean, complete scans of isolated plants [8, 14], and methods that tolerate field occlusion complete unstructured geometry instead [27].

In this work, we present CROPCODE, a framework that makes this inverse problem tractable by using the procedural generator itself as the prior. We pretrain a four-stream masked autoencoder [3, 10, 21, 32] over RGB, depth, point clouds, and tokenized procedural parameters of 15,000 generated plants per species, maize and sorghum, and fine-tune it on simulated rows in which neighboring leaves bend on contact. Because random modality subsets are masked during pretraining, a single model learns to complete any subset at inference. A full-modality teacher is then distilled [22] into single-sensor students for deployment. Distillation lowers the cross-modal generation loss by 30% in sorghum and 44% in maize, and it halves the sorghum parameter error from a point cloud alone. Row fine-tuning against the target plant behind its neighbors removes most of the error that the neighbors add: 92% in maize and 85% in sorghum.

We also address a design question that arises for any model with access to generative structure: where should the parameter stream enter? In our controlled ablations, a parameter stream that is largely visible to the encoder harms the representation in both species. With the shared token budget, adding the parameter stream to the encoder degrades the frozen representation on every sorghum phenotype target, and on maize the paired comparison is a draw. Control arms that restore the sensor-token budget separate the species: in sorghum the loss persists, even when the parameters serve only as a reconstruction target for the decoder, while in maize the arms that see 20 % of the parameters or only reconstruct them improve every trait. Supervising parameters as a generation target, however, lets the distilled generalist outperform a parameter-only specialist at parameter regression (first-release result: no final-data specialist was trained). One interpretation, consistent with modality competition [35] and negative transfer [49], is that a stream that already contains the answer suppresses learning in the streams that must provide the trait information at test time.

Adding depth to the photograph and the point cloud lowers Chamfer distance by 28% in sorghum and 25% in maize, whereas a linear probe on the frozen representation, read out on phenotypic traits, finds little or no gain from depth and identifies RGB as the main source of trait information. On slender geometry the two metrics measure different quan-

tities, surface proximity and recovered structure, and we adopt the probe as the deciding metric for modality decisions [6, 48].

We claim that fragmentary field sensing suffices to recover a complete, interpretable procedural plant, provided the generator is used for both the training data and the output representation. We validate this claim on real field scans of maize and sorghum with independently reconstructed ground truth [18, 25], including rows in which neighbors occlude the target, demonstrating accuracy matching staged annotation pipelines while running in milliseconds per plant. The key contributions of this work are:

1. **Task and framework.** The first formulation and system, to our knowledge, for recovering procedural plant representations of slender, row-occluded field crops from *any subset* of RGB, depth, and point-cloud input in one amortized forward pass, with full-modal to single-sensor distillation (generation loss $-30\%$ in sorghum and $-44\%$ in maize; sorghum parameter error from a point cloud alone halved), and row fine-tuning that removes most of the error added by neighboring plants (92% in maize, 85% in sorghum).
2. **A design rule for structured modalities.** Controlled, budget-matched, two-species evidence that a parameter stream largely visible to the encoder degrades the sensor representation in both species, while the value of the parameters as a reconstruction target depends on the generator (it degrades the sorghum representation and improves the maize one); we expect this guidance to transfer to any multimodal model with access to privileged generative structure.
3. **An evaluation protocol for slender geometry.** A demonstration that Chamfer distance and downstream phenotype probes disagree systematically on slender organs, and a frozen-latent probe protocol that resolves modality decisions where Chamfer is insensitive.
4. **A benchmark.** FIELDPROC, a two-species procedural field benchmark of 30,000 plants spanning diverse phenotypes, generator configurations, and row occlusion severities, with paired multimodal captures, physically simulated row occlusion, and exact parameter ground truth, together with a real-field evaluation split carrying reconstructed procedural ground truth across the genotypes of a diversity panel [25]. No existing dataset combines these elements (Tab. 2) [42, 52]. We will release the data, generation code, and evaluation scripts.

## 2. Related Work

**Recovering procedural plants.** Inverse procedural modeling has a long history for woody plants: parameters of tree generators have been recovered by stochastic search [44], by Markov chain Monte Carlo inference [45], and by inferring L-systems from point clouds [17, 28]. These methods optimize or infer per instance. Recent systems address crops specifically. Demeter learns a parametric morphology model from soybean point clouds [8], Ghrer et al. infer grammar-style plant representations from 3D scans [14], CropCraft fits a procedural crop model by Bayesian optimization against multi-view imagery [53], and a recent staged pipeline reconstructs procedural maize from terrestrial laser scans through vision-language-model annotation and differentiable NURBS refinement [9, 18]. Each of these methods presupposes a single rich observation, a clean and complete point cloud or a multi-view capture of an isolated plant, and none addresses missing modalities or inter-plant occlusion.

**Programs from observations.** Outside agriculture, decoding parametric programs from observations is an established pattern: neural shape programs [46], diffusion-based inverse procedural content generation from a single image [54], text-conditioned CAD generation [24], and cadrille, which maps images, point clouds, or text to executable CAD models [26]. These systems operate on rigid objects observed cleanly and in isolation. Our setting differs in the geometry (deformable, slender organs), in the input (noisy, occluded field sensors), and in a design question that none of them raises: whether the program parameters should be visible to the encoder at all.

**Completion and slender geometry.** Learned point cloud completion has progressed from folding-based decoders [52] to geometry-aware transformers [5, 50, 51], trained on benchmarks that contain compact, largely convex objects and no slender-under-occlusion category. Completion of thin plant organs under occlusion has emerged recently [27], and reconstruction of thin structures from clean multi-view data is an active topic [13, 39], but in both lines the output is unstructured geometry rather than an editable model. Chamfer distance, the standard completion metric, is insensitive to thin sheets and can be reduced by collapsing predictions onto bulk structure [30, 43, 48]; we therefore evaluate with trait probes alongside geometric metrics (Sec. 4.7).

**Multimodal masked pretraining.** Masked autoencoding [21] extends to multiple modalities in MultiMAE [3], to any-to-any generation over tokenized modalities in 4M [4, 32], to point clouds in Point-MAE [34], and to combined RGB, depth, and point cloud inputs in EmbodiedMAE, on which our architecture is based [10]. Robustness to missing modalities is a recognized concern [41], and distillation from multi-sensor teachers to single-sensor students is established for discriminative tasks [22]. We extend these methods in two ways: we add a structured generative modality, encoded with continuous number tokens [15], and

we distill a generative full-modality teacher for structured parameter regression.

**Learning from procedural data.** Pretraining on purely synthetic data is viable, from abstract fractals [23] to procedurally generated scenes [16, 40], with domain randomization reducing the gap to real data [47] and with the value of synthetic data known to depend on the downstream task [11]. Our setting is the favorable case for this strategy: the procedural generator defines the output representation, spans the morphology of the target crop, and is paired with real field evaluation data [25]. Our results add one caveat to this strategy: the generator's parameters are valuable as data, but exposing most of them to the encoder reduces representation quality in both species, and their value as a reconstruction target depends on the generator (Sec. 4.6).

## 3. Method

In this section, we present CROPCODE, a multimodal framework for recovering procedural plant representations from incomplete sensor observations. We first formulate the reconstruction task and describe the procedural data generation process. Next, we introduce the token representations and masked pretraining objective. Finally, we describe the teacher–student training strategy for single-sensor inference and the protocol used to evaluate transfer to field observations. An overview of the framework is shown in Fig. 2.

### 3.1. Problem formulation

Let $G$ denote a procedural generator that maps a parameter vector $\theta \in \mathbb{R}^{d_\theta}$ to a plant mesh $G(\theta)$. The parameters describe plant-level structure, including stem and panicle geometry, together with leaf-level attributes such as insertion height, length, roll, branching angle, and blade width. We consider three sensor modalities, $\mathcal{M} = \{\mathrm{rgb}, \mathrm{d}, \mathrm{pc}\}$, corresponding to an RGB image, a depth frame, and a point cloud. In field conditions, only a subset $S \subseteq \mathcal{M}$ may be available, and each observation may contain partial geometry or occlusion from neighboring plants. Our objective is to recover the underlying procedural representation from these observations.

We formulate this task as feed-forward prediction:

$$f_\phi(\{x^m : m \in S\}) = \left(\hat{\theta}, \{\hat{x}^m : m \in \mathcal{M}\}\right), \qquad (1)$$

where $\phi$ denotes the network parameters, $\hat{\theta}$ is the predicted plant description, and $\hat{x}^m$ denotes the reconstructed sensor modalities. The procedural output $G(\hat{\theta})$ supports geometric reconstruction and trait analysis within the same representation. Per-instance fitting of a procedural model to a scan [18] solves an optimization for every plant; we instead learn a shared inverse mapping that is applied to each new observation in a single forward pass.

| | Maize | Sorghum |
|---|---|---|
| Plants | 15,000 | 15,000 |
| Split (train/val/test) | 10,500 / 2,250 / 2,250 | |
| Views per plant | 10, random | 10, fixed rig |
| Camera distance | fitted per plant | 2.5 m |
| Per-view cloud | 8,192 pts, visible | complete |
| Parameter slots ($d_\theta$) | $14 + 14 \times 28$ | $9 + 5 \times 24$ |
| Leaf count range | 3–22 | 4–35 |
| Plant height | 0.62–1.44 m | stem 0.19–1.74 m |
| Rows × orientations | 15,000 × 10 | 15,000 × 10 |
| Row camera | level, 1.0 m | level, 2.5 m |
| Row cloud in / target | 24,576 / 8,192 pts | |
| Real-field split | MaizeField3D | TBD |

Table 1. FIELDPROC dataset configuration. The parameter dimension includes plant-level attributes and a fixed number of leaf slots. Every row has one main plant from the split and two neighbors at a 0.15 m stem spacing, widened only when the contact solver needs it (Supp. C). Confirm the real-data splits [25].

The procedural generator provides both the training observations and their exact parameter targets. This allows us to supervise parameter recovery without manually annotating each synthetic plant. At deployment, however, $\theta$ is unknown and must be inferred exclusively from sensor input. We therefore distinguish its role as a reconstruction target from its possible use as privileged input during training. The visibility of the parameter stream is controlled separately, allowing us to study this distinction while keeping the sensor-token budget fixed (Sec. 4.6).

### 3.2. A procedural field-data engine

**Procedural generation.** We construct FIELDPROC using species-specific generators for maize and sorghum, 15,000 plants each. Plant $i$ is generated with random seed $i$ and is paired with its exact procedural parameters. Maize traits are drawn from Gaussians (for example, height $0.83 \pm 0.10$ m, leaf count $12.4 \pm 2.6$, leaf angle $44^\circ \pm 10^\circ$) coupled through four latent traits (vigor, slenderness, posture, texture), with canopy-position curves and per-leaf jitter; the plants are 0.62–1.44 m tall. Sorghum plants come from the EvoEngine sorghum generator, which draws the heading, stem tilt, leaf count and internode length per plant and evaluates per-leaf curves for insertion point, length, width, roll (alternating sides), branching angle and bending; the sorghum plants have 4–35 leaves (mean 19.5) on stems of 0.19–1.74 m. The populations therefore span continuous variations in morphology rather than discrete genotype classes, and we evaluate phenotype representations through regression. The generators also differ in their trait dependencies. In sorghum, stem length is the internode length times the leaf count, so it is strongly coupled to leaf count (stem length $= 0.05\, n_{\text{leaves}} - 0.001$, $R^2 = 0.988$), whereas

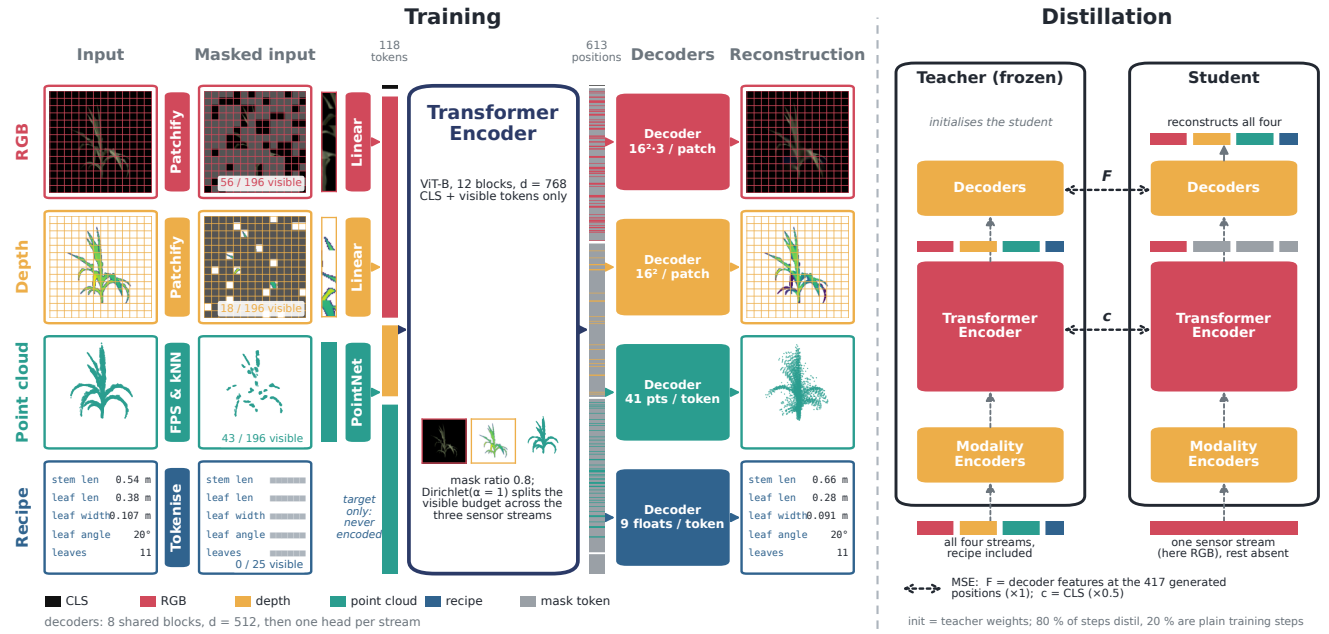



Figure 2. **CropCode overview.** (a) Four token streams enter a shared encoder under a Dirichlet masking budget; modality-specific decoders reconstruct all streams. During pretraining the parameter stream is both masked input and decoder target; its role is ablated in Sec. 4.6. (b) The full-modality teacher is distilled into single-sensor students for deployment.

this association is weak in maize ($r = 0.20$). We therefore evaluate both species to examine whether the observed behavior extends beyond a single generator's correlations.

**Isolated-plant observations.** For each plant, we render ten views with `pyrender` at $1024 \times 1024$ and a $40°$ vertical field of view, each with an RGB image, a depth image stored losslessly as a packed 32-bit value, and the camera pose. Three fixed directional lights are aimed at the plant's bounding-box center. Images are rendered on a gray background; in the final release the background is made transparent and composited over black. Sorghum views follow one fixed Fibonacci-sphere rig at 2.5 m, five directions above and five below the horizon (elevations $-64°$ to $+64°$), so every plant is seen from the same ten directions, and the per-view cloud is the complete, occlusion-free plant cloud expressed in the camera frame. Maize views are ten directions drawn uniformly on the sphere per plant, at a distance fitted to the plant's bounding box, with an extra light at the camera; the per-view cloud is 8,192 points back-projected from that view's depth image, so it holds only the surface visible to the camera. At 2.5 m the lens covers about 1.8 m of height, so the top or base of the tallest sorghum plants can fall outside the frame.

**Row observations.** In the row sets, each plant of a split is placed, as the *main* plant, between two neighbors from the same split (one row per plant), with 0.15 m between stem bases. Each row is rendered at ten orientations: the main plant is turned about its stem by $0°, 10°, \ldots, 90°$ and each neighbor by the same angle plus a uniform offset in $\pm 15°$; sorghum neighbors are also tilted so their stems run parallel to the main stem. Before rendering, a contact solver makes the three plants free of intersections while keeping them close to their grown shapes: each leaf blade is a chain of 8 cm segments whose joints bend along circular arcs, so blades stay smooth, keep their length and stay attached, while stems, sheaths and tassels stay rigid. A leaf that would cross another plant bends by the smallest clearing angle at the joint before the contact; neighbors bend first, and a main-plant leaf bends only where a neighbor cannot move. Contacts are decided by exact triangle–triangle tests, so leaves may touch but never intersect. When no layout exists under the bend limits, the limits are loosened in stages and then the spacing is widened; in maize, 76–79 % of orientations are solved at 0.15 m and none requires more than 0.33 m. A rover camera, level and aimed at the main plant's bounding-box center from 1.0 m (maize) or 2.5 m (sorghum), takes one shot per orientation, rendered twice: the input with all three plants (RGBA, depth, and a 24,576-point cloud of every visible plant) and the ground truth with the main plant alone, in its bent shape, from the same camera (8,192 points). The parameter record describes the grown plant, so the target geometry departs from it only

where the solver had to bend a main-plant leaf. Occlusion is measured per shot from the two depth maps (maize, 120 train rows): the fraction of the main plant's own pixels hidden behind a neighbor is 9 % on average, 13 % at 0°, where the generator's single leaf plane lies along the row, falling to 1 % at 90° (90th percentile 17 %), and the neighbors account for 59–68 % of the plant pixels in the frame and two thirds of the input cloud. Sorghum leaves spread in every direction, so its occlusion does not depend on yaw: on the 22,500 test rows, 23 % of the main plant's pixels are hidden on average at every yaw (90th percentile 34 %), and the neighbors account for 57 % of the plant pixels in the frame. On the maize test rows the same measures reproduce the training-row values above (9 %, 13 % at 0°, 1 % at 90°).

**Data partitioning.** We partition plants into training, validation, and test sets in a 70/15/15 ratio by plant, so all views and rows of a main plant fall in one split, and we bias the held-out sets toward unusual morphologies. For maize, each plant is scored by its Mahalanobis distance [31] from the median over 19 robustly standardized parameter features, and the most extreme plants are assigned alternately to test and validation, a pure tail split. For sorghum, the extremeness is the Mahalanobis distance from the mean over six features (leaf count, stem length, mean and maximum leaf length, mean and standard deviation of leaf roll), and the probability of being held out rises steadily with a plant's extremeness rank, from 0.3 % in the least extreme 5 % of plants to 70 % in the most extreme 5 %; validation and test share one distribution. The sorghum selection rule is inferred from the released assignment; confirm it against the script that wrote it. We report errors in native trait units alongside $R^2$, since the latter depends on the target distribution of each split. In the ablations, an epoch samples one view per training plant, with identical view draws across compared models. Data-scaling experiments use nested subsets of plants, each retaining all ten available views, and match the number of optimizer updates.

### 3.3. Four-stream masked pretraining

**Token representations and architecture.** Our architecture builds on EmbodiedMAE [10], extending its sensor streams with a structured parameter representation. RGB and depth are each represented by 196 patch tokens on a $14 \times 14$ grid, with fixed 2-D sine–cosine positional embeddings. Following Point-MAE [34], we select 196 point-cloud centers through farthest-point sampling and group their 32 nearest neighbors. A local PointNet [37] embeds each group using coordinates relative to its center. The parameter stream consists of $1+K$ tokens: one for plant-level attributes and $K$ for individual leaves, with $K = 24$ for sorghum and $K = 28$ for maize. Each attribute is affinely normalized to $[0, 1]$ and linearly projected, retaining continuous numerical values rather than text tokens, in the spirit of xVal [15]. A validity mask identifies padded leaf slots. Learned modality embeddings distinguish the streams, and a CLS token is prepended to the visible sequence. A 12-block encoder of width 768 processes this sequence, followed by an 8-block decoder of width 512. The decoder restores a fixed ordering of all $L = 588 + (1 + K)$ token positions using mask tokens for missing inputs. Linear heads reconstruct RGB and depth patches, a FoldingNet-style head generates 41 points per point-cloud token, and an MLP predicts the procedural parameters. Plants with more than $K$ leaves keep their first $K$ leaf tokens; this affects 1,586 of the 15,000 sorghum plants (10.6 %) and no maize plant. The validity mask enters only the loss: the model is not trained to predict which leaf slots are occupied, and every leaf metric in this paper is scored on the real leaf slots of the ground truth. State how the leaf count is chosen when a predicted recipe is re-rendered.

**Modality masking.** We use a total mask ratio of $\rho = 0.80$ over all four streams, giving a visible budget

$$V = \left\lfloor (1-\rho) \sum_{m \in \mathcal{M}} L_m \right\rfloor, \tag{2}$$

which is 122 tokens for sorghum ($L = 613$) and 123 for maize ($L = 617$). At each training step, this budget is allocated across the four streams, the parameter stream included, by a sample from $\mathrm{Dirichlet}(\mathbf{1})$. The allocation is rounded to integers that sum to the budget, with at least one visible token and at least 25% masked tokens per stream. Token counts are shared across a batch, while visible positions are sampled independently for each example. The procedural parameters are therefore both an encoder input and a decoder target in the headline model: because the parameter stream is short, it is on average ∼58% visible, while the three sensor streams together keep about 107 of their 588 tokens.

For the controlled ablations of Sec. 4.6, the parameter stream can instead be taken out of the shared budget and masked at its own ratio $\rho_\theta$:

$$V_\theta = \begin{cases} 0, & \rho_\theta = 1, \\ \max\left(1, \lfloor (1+K)(1-\rho_\theta) \rfloor\right), & 0 \le \rho_\theta < 1, \end{cases} \tag{3}$$

with the Dirichlet budget then allocated over the three sensor streams alone ($V = 117$). Allocating the budget over the sensor streams alone keeps the sensor masks identical across matched three- and four-stream arms under the same seed. $\rho_\theta = 1$ makes the parameters a reconstruction target only, never an encoder input. Parameters are never an input at deployment, where only sensor streams are available.

| | Real | Slender | Row occl. | RGB+D+PC | Proc. GT | Scale |
|---|---|---|---|---|---|---|
| PCN / ShapeNet-55 [50, 52] | × | × | × | × | × | 55K |
| Pheno4D [42] | ✓ | ✓ | × | × | × | 84 |
| Crops3D [55] | ✓ | ✓ | × | × | × | |
| MaizeField3D [25] | ✓ | ✓ | × | × | (✓) | 1K |
| FIELDPROC (ours) | ✓† | ✓ | ✓ | ✓ | ✓ | N |

Table 2. Dataset comparison. Proc. GT: exact generative parameters per plant; (✓): fitted post hoc rather than exact. †real-field evaluation split; training data is procedural. verify Crops3D scale; consider adding A-Occ-Plant's data if released

**Reconstruction objective.** We jointly optimize the four reconstruction terms:

$$\mathcal{L}_{\text{rec}} = \ell_{\text{rgb}} + \ell_{\text{d}} + \ell_{\text{pc}} + 5\ell_{\theta}. \tag{4}$$

The RGB loss is MSE on patch-normalized targets [21], while the depth loss is MSE after per-image min–max normalization. Both are evaluated at masked positions during pretraining. The parameter loss is Smooth-L1 over masked, valid parameter tokens. For point clouds, the loss compares the full predicted set $\hat{P}$ with the target set $P$. We use a sigmoid-weighted Chamfer objective, motivated by the limitations of nearest-neighbor geometry objectives [43, 48]:

$$\begin{aligned}\ell_{\text{pc}} = &\frac{1}{|\hat{P}|}\sum_{p\in\hat{P}} w(\delta(p,P))\,\delta(p,P) \\ &+ \frac{1}{|P|}\sum_{q\in P} w(\delta(q,\hat{P}))\,\delta(q,\hat{P}),\end{aligned} \tag{5}$$

where $\delta(p,P) = \min_{q\in P}\|p-q\|_2$ and $w(r) = \sigma(a(r-\tau))$. We set $a = 100$ and $\tau = 0.01$, reducing the weight of small residuals relative to larger reconstruction errors. Both directions are averaged over their own set and then over the batch. Clouds are centered on their centroid and scaled so the farthest point lies on the unit sphere, so $\tau$ is relative to the plant: one percent of its radius, about 7 mm on a 1.4 m plant. Nearest-neighbor losses let many predicted points share one target point without penalty, so the folding head can collapse a token's points into a clump and still score well. For the headline model, we therefore add an optimal-transport term that matches the two clouds as distributions, the debiased Sinkhorn divergence [12],

$$\ell^{\star}_{\text{pc}} = \ell_{\text{pc}} + 0.5\, S_{\varepsilon}\left(\hat{P}_{1024}, P_{1024}\right), \tag{6}$$

with $p$=2 and blur $\varepsilon$=0.02 in the unit-sphere frame, computed between random 1,024-point subsets of prediction and target that are redrawn at every step. The term rewards coverage of the whole cloud, and the subsets keep its cost matrix at $B \times 1024^2$. The ablation arms of Sec. 4 were trained with Eq. (5) alone; the Sinkhorn term is used only for the headline model and the distillation. Report the ablation of this term (E7). We train with AdamW, $\beta = (0.9, 0.95)$, weight decay 0.05, linear warm-up, cosine learning-rate decay, and gradient clipping at 1.0. The main schedule uses 1,000 epochs and a global batch size of 256. Ablations instead use a fixed budget of 197,400 optimizer steps at global batch size 32 to prevent larger datasets from receiving more updates merely through equal-epoch training. The main schedule visits all ten views of every training plant in each epoch, with a peak learning rate of $6 \times 10^{-4}$ and a 20-epoch warm-up. Ablations sample one view per plant per epoch, with a peak learning rate of $1.5 \times 10^{-4}$ and a warm-up of 40 epochs at the full-data scale (scaled with the epoch count when the number of plants changes).

### 3.4. A full-modality teacher, single-sensor students

During pretraining, every sensor stream keeps some visible tokens. Consequently, single-sensor inference introduces a missing-modality configuration that the encoder has not encountered during this stage. We address this gap through teacher–student distillation [22]. The teacher is the pretrained four-stream model, frozen: for sorghum the final checkpoint (epoch 1,000), and for maize the final epoch-600 checkpoint. It receives RGB, depth, point clouds, and the procedural parameters, the same four streams it was pretrained on, so the parameters act as privileged information that the students will not have. For sorghum every token of every stream is visible to the teacher; for maize we retain half the tokens of each stream to avoid the reconstruction degradation observed with fully visible input. We assess the teacher's reconstruction quality under the selected mask before distillation. The teacher provides a CLS representation $c^T \in \mathbb{R}^{768}$ and final normalized decoder features $F^T \in \mathbb{R}^{L\times 512}$, where $L = 613$ for sorghum and 617 for maize. The fixed decoder layout gives a one-to-one correspondence between teacher and student features at every output position.

The student is initialized from the teacher and optimized for 100 epochs with AdamW, a learning rate of $10^{-4}$, cosine decay, and global batch size 128. At each step, a seeded random draw selects single-sensor training with probability 0.8 or ordinary masked pretraining with probability 0.2. In a single-source step, the source $s$ is cycled deterministically over all four streams (RGB, depth, point cloud, and

parameters), and only its tokens enter the encoder; optionally, half these tokens are also masked for regularization. For the two-stream students, $s$ is a pair of streams whose tokens enter the encoder together, and only the point cloud is reconstructed. The objective combines ground-truth reconstruction with decoder- and encoder-level feature matching:

$$\mathcal{L}_{\text{student}} = \mathcal{L}_{\text{rec}} + \frac{\lambda_F}{|\mathcal{J}_{\neg s}|} \sum_{j \in \mathcal{J}_{\neg s}} \|F_j^S - F_j^T\|_2^2 + \lambda_c \|c^S - c^T\|_2^2, \tag{7}$$

where $\mathcal{J}_{\neg s}$ contains the decoder positions belonging to non-source streams, including the procedural parameters. We use $\lambda_F = 1$ and $\lambda_c = 0.5$. Through these terms, the outputs that the student must infer from the available sensor are supervised by the teacher's privileged information. The student does see parameter tokens during training: on the parameter-source steps, which train it to regenerate the plant from its program alone, and on the interleaved ordinary masked steps, which use the four-stream pretraining mask and maintain reconstruction across modalities. Specialists restrict the source modalities and, optionally, the reconstruction targets, whereas the generalist cycles across all four sources. At deployment only a sensor source is available; we evaluate each source alone, without source-token masking, and compare against the initialization using the same evaluation routine. On a single-sensor step, the RGB, depth and parameter losses are averaged over masked tokens only, so a fully visible source stream contributes nothing to $\mathcal{L}_{\text{rec}}$; when half of the source tokens are masked, those masked tokens do. The point-cloud loss is always taken over the whole cloud, so a point-cloud source is still reconstructed. The feature terms of Eq. (7) exclude the source positions for the generalist, and for a specialist they cover only its target streams.

### 3.5. Crossing into the field

1-2 paragraphs

We evaluate both procedural reconstruction and the trait information retained in the learned representation. For phenotype analysis, we fit ridge regressors to the frozen encoder's CLS features using the sensor streams available to each experimental configuration. Procedural parameter inputs are excluded, and the parameter tensor is zeroed as a safeguard against target leakage. Each plant contributes one deterministic view, and feature standardization is fit on the training split only. Performance is measured on the held-out plants with paired bootstrap confidence intervals over plants. We regard a modality as improving a trait only when the confidence interval for the paired performance difference excludes zero in the favorable direction on both validation and test sets. To assess partial observations, we ray-cast the plant mesh from the rendering viewpoints and construct one-camera and three-camera clouds in addition to complete clouds. We evaluate probes both by refitting under each visibility condition and by transferring a probe trained on complete clouds. Comparing the two protocols separates recoverable trait information from sensitivity to a change in input coverage. Partial-cloud reconstruction uses masked input to remain closer to the pretraining regime.

**Row protocol.** Models are evaluated on the rows in two ways. *Zero-shot*: the isolated-plant model is fed the row observation as it is. *Row fine-tuning*: the pretraining objective is kept but its target changes. The encoder sees the masked row streams (RGB, depth, the scene cloud with both neighbors in it, the recipe in its usual budget) and the decoders are supervised with the main plant alone, its RGB, depth and cloud placed in the scene cloud's frame, and its recipe, so the model learns to reconstruct the plant behind its neighbors rather than to copy the scene. Training uses one orientation per plant per epoch and runs 100 epochs at a learning rate of $10^{-4}$ and global batch 32, starting from the isolated checkpoint. The control arm runs the same schedule on the isolated twins. In the evaluation (Sec. 4.4), every held-out item is scored under both inputs, with the recipe never an input: Chamfer of the generated cloud against the main plant *and* against the input cloud (a model that merely echoes the scene scores well on the second and badly on the first), recipe error on the real parameter columns, both by orientation and by occlusion quartile, and the paired row-minus-isolated difference per plant with a bootstrap interval.

For real-field evaluation, procedural reference models will be obtained independently from scan-based reconstruction, using the maize pipeline and dataset described in [18, 25]. We will compare predicted parameters and re-rendered geometry with these reconstructed references. Unlike synthetic parameter labels, these references are fitted estimates and may contain errors from incomplete scans. The evaluation will include the effect of neighboring-plant occlusion and will compare real-data transfer with the controlled partial-cloud experiments. The partial-cloud experiments isolate a visibility shift; they do not by themselves quantify all sources of the simulation-to-real gap. Finalize real-data splits and species coverage, including sorghum reference provenance, plant segmentation/matting, camera-frame alignment, row-plot inclusion, and treatment of unobserved leaves.

## 4. Experiments

### 4.1. Setup

**Models and budgets.** All ablations share one budget of 197,400 optimizer steps at global batch 32 (600 epochs of one view per plant over the 10,500 training plants), one view-sampling seed, and the base ViT (114 M parameters)

unless model size is the variable; arms that change the number of training plants keep the step budget by changing their epoch count (Sec. 3.3). The headline model differs in schedule (1,000 epochs over all ten views at global batch 256) and in its point-cloud loss, which adds the Sinkhorn term of Sec. 3.3. The headline model, like the four-stream ablation arm, takes the parameter stream as an encoder input under the shared masking budget, and the teacher and every student of Sec. 4.5 are initialized from its weights. The sorghum headline model and its distillation are on the final release; the few headline-derived numbers still from the first-release, Chamfer-only run are marked where they appear. A maize headline model on the same schedule and loss reached 1,000 epochs (validation Chamfer $0.67 \times 10^{-3}$); its generalist, the three RGB students and two of the four two-stream students are distilled, and the other two are training.

**Metrics.** We measure geometry by the symmetric Chamfer distance in the unit-sphere frame ($\times 10^{-3}$; a plant spans $\approx$1.4 m, so 1.0 is about 7 mm), computed on the masked validation step for pretrained arms and with the source unmasked for students. Parameter error is the mean absolute error on the normalized $[0, 1]$ fields of hidden, non-padding leaf tokens; where a per-field number is more telling we report *skill*, $1 - \mathrm{MAE}/\mathrm{MAE}_{\mathrm{mean}}$, so that 1 is exact and $\leq 0$ is no better than the average plant. Phenotype readout follows Sec. 3.5: ridge regression on the frozen CLS token with the parameter tensor zeroed, one deterministic view per plant, reported as $R^2$ on the extreme-enriched held-out plants (2,250 val, 2,250 test; val unless stated) with paired bootstrap intervals over plants (2,000 resamples). A difference is a *win* only when its interval excludes zero on val *and* test. Sorghum's four named traits have rank two by construction (height and leaf count are one generator variable, $r$=0.994, and the biomass proxy is that variable again), so we add the independent axes varied by the generator, leaf roll and leaf length. Maize has eleven largely independent traits (Sec. 3.2). Implementation details, per-trait tables and intervals are in the supplement.

### 4.2. Regime I: single-view reconstruction of isolated plants

Table 3 covers deployment on an isolated plant: from one photograph, one depth frame or one cloud, the model returns the recipe and the other streams in one forward pass. On sorghum, the cloud generated from a single photograph has Chamfer 0.88 (test 0.79), the cloud generated from a depth frame has 0.62, and the cloud regenerated from the input cloud has 1.22. The full-modality teacher reaches 0.35. The photograph-to-cloud student has skill 0.984 against the mean plant of the split. On the first-release leaf fields, a cloud gives skill 0.95 for insertion height, 0.93 for branching angle, 0.67 for leaf length and 0.50 for roll, and a photograph gives 0.90, 0.85, 0.66 and 0.33. The cloud is ahead of the photograph on every field with nonzero skill, and the gap is widest on roll. The three waviness fields of that release were unrecoverable from any source (skill $-0.04$ to 0.02): the head emitted the population mean, and a single aggregate MAE would have averaged these constant predictions with the fields that have skill. The final generator does not include these fields. Fig. 3 shows one held-out plant per species generated from the photograph alone: the headline model used as is produces a diffuse cloud around the right silhouette, the distilled generalist recovers the individual blades, and the leaf lengths it reads off the photograph follow the ground truth along the stem. Add the edit demo: change one recovered parameter and re-render.

### 4.3. Regime II: morphological complexity, viewpoint, self-occlusion

**Viewpoint.** The ten cameras of a plant form an elevation ladder ($\sin \mathrm{elev} = -0.9 + 0.2\,i$), and every view shares one cloud and one recipe, so the view index changes the input but not the target. Figure 4A shows the photograph-to-cloud error by elevation. The error is U-shaped in elevation, lowest for side-on views and highest for views looking steeply down *or* up (top-down views are hard, but so are bottom-up ones). The curve has the same shape for the pretrained model and every student. Distillation lowers the error by 28 % on average and improves nine of ten views, but does not flatten the curve (best-to-worst ratio 2.35 $\rightarrow$ 2.49). Across the 2,250 test plants the camera's distance from the horizon explains $\eta^2 = 0.24$–$0.31$ of the per-plant error, and no measure of the plant itself (leaf count, stem length, slenderness or extent of the cloud) correlates beyond $|\rho| = 0.06$. In this regime, the view rather than the morphology determines how accurately a photograph can be inverted. The recipe is the more view-stable output: the recovered insertion heights of one plant vary by 0.08 of the between-leaf spread across its ten photographs, and the recovered branching angles vary by 0.12. Distillation halves the error of both fields ($-59$ and $-60$ %). The zero-skill waviness fields did not improve: their error stayed flat and their view spread grew from 0.18 to 0.26, so a field that no input can predict is not neutral, because it becomes more view-dependent under distillation.

**Training views.** We restricted the pretraining draw of the four-stream arm to a band of the elevation ladder at equal steps: views 0–1 (top-down), views 4–5 (near the horizon, the height of a tractor-mounted camera) or view 0 alone, against all ten views. The probe scores view 0 for every arm. Paired over plants against the all-view arm, the horizon arm loses six of the nine learnable targets on both splits (test: roll $-0.065$, leaf angle $-0.024$, size $-0.010$ to $-0.013$). The top-down arm wins three targets and ties on

| input | leaf cnt (acc±1) ↑ | height (rel. err) ↓ | leaf params (mean skill) ↑ | F-score ↑ | Chamfer $\times 10^{-3}$ ↓ |
|---|---|---|---|---|---|
| *Sorghum* | | | | | |
| RGB (1 view) | | | 0.69* | | 0.88 |
| RGB (3 views) | | | | | |
| depth (1 view) | | | | | 0.62 |
| PC (1 view) | | | 0.76* | | 1.22 |
| RGB+D+PC | | | | | 0.35 |
| two-stage (completion→fit) | | | | | |
| *Maize* | | | | | |
| RGB (1 view) | | | | | 0.98 |
| RGB (3 views) | | | | | |
| depth (1 view) | | | | | 0.99 |
| PC (1 view) | | | | | 1.97 |
| RGB+D+PC | | | | | |
| two-stage (completion→fit) | | | | | |

Table 3. **Regime I: isolated plants, one view.** Each row is the distilled generalist given that input alone; the RGB+D+PC row is the teacher. Before/after distillation is in Tab. 6. Chamfer against the complete plant cloud, val. *Mean skill over the four leaf fields with any skill (insertion height, branching angle, length, roll), on an 8-plant sample of the first release; the three waviness fields of that release have zero skill from every source and were dropped from the final generator. Fill leaf-count accuracy, height error, F-score and the 3-view and two-stage rows from the FIELDPROC evaluation script; maize numbers are from the 55-epoch first-release student.

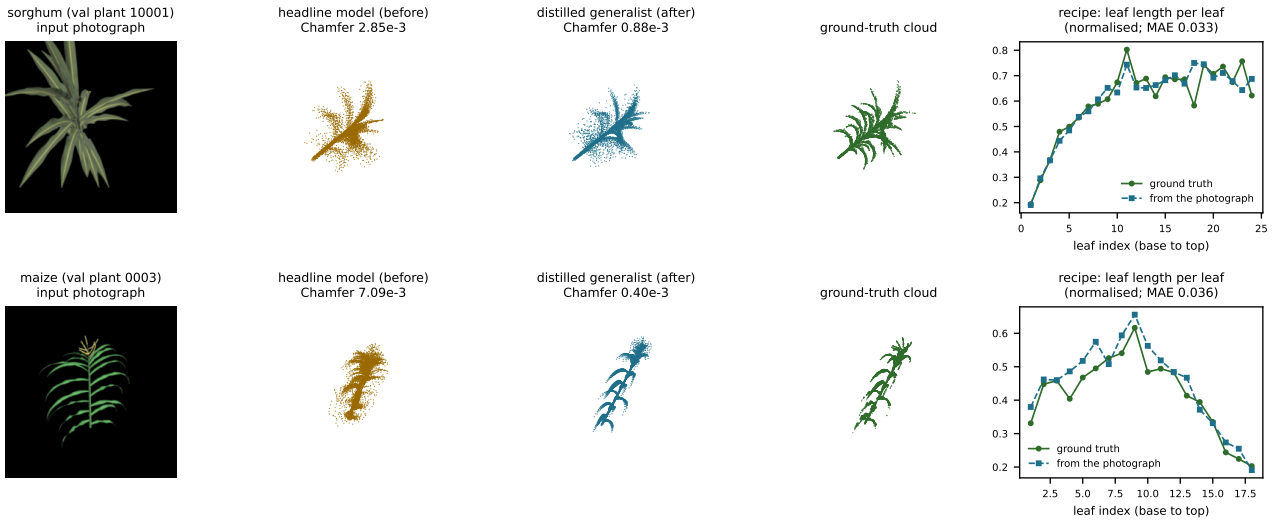



Figure 3. **From one photograph to a plant (held-out plants).** Each row takes only the photograph on the left. The headline model before distillation and the distilled generalist generate the point cloud; titles give the Chamfer distance to the ground-truth cloud. The right panel compares the generalist's recovered leaf lengths (normalized) with the generator's, leaf by leaf from the base.

size, and the single-view arm wins six (test: roll $+0.061$, leaf length $+0.069$, size $+0.008$). View 0 lies inside the training band of the top-down and single-view arms and outside the band of the horizon arm, so these differences measure the overlap between training and test views as much as the breadth of the training draw. To separate the two, we refit and scored the probe on each view $k$ of every plant (Fig. 5). Each restricted arm leads on roll only on the views it trained on: against the all-view arm, test roll is $+0.061$ (single), $+0.042$ (top-down) and $+0.027$ (horizon) on the training views and $-0.066$, $-0.059$ and $-0.054$ on average on the other views, down to $-0.144$ for the single-view arm on view 6. The all-view arm is the most even, $0.734$ to $0.778$ across the ten views. Size separates the arms much less: height stays within $0.02$ of the all-view arm on every view, and the restricted arms lose most on the steepest views (horizon $-0.013$ on view 0; top-down $-0.019$ and horizon $-0.014$ on view 9, which looks up at the plant). A narrow band of training views therefore raises the probe on that band and lowers it elsewhere, so a deployment camera whose elevation is known in advance favors training at that elevation, and an unknown one favors the full ladder.

| | matched probe, height | | | transferred probe, height | | | Chamfer $\times 10^{-3}$ | | |
|---|---|---|---|---|---|---|---|---|---|
| cloud coverage | full 100 % | 3 cam 78 % | 1 cam 49 % | full | 3 cam | 1 cam | full | 3 cam | 1 cam |
| PC | 0.888 | 0.839 | 0.765 | 0.888 | 0.509 | 0.575 | 4.99 | 6.18 | 12.8 |
| PC+RGB | 0.977 | 0.967 | 0.948 | 0.977 | 0.624 | 0.001 | 2.49 | 3.64 | 11.5 |
| PC+RGB+D | 0.975 | 0.968 | 0.956 | 0.975 | 0.721 | 0.576 | 1.76 | 3.06 | 11.4 |
| four streams | 0.961 | 0.953 | 0.946 | 0.961 | 0.748 | 0.818 | 2.52 | 3.62 | 11.4 |
| headline (E1) | **0.982** | **0.975** | **0.960** | **0.982** | 0.623 | −0.26 | **1.36** | 3.14 | 11.6 |

Table 4. **Self-occlusion: what one or three cameras see (sorghum, test).** Every model was trained on complete clouds; at evaluation the cloud is cut to the points visible from camera 0, or cameras 0–2, by ray-casting against the mesh. Matched: probe refit per condition. Transferred: the full-cloud probe applied as is. Chamfer against the complete cloud, masked regime, val. Final data release; the headline model at epoch 1,000.

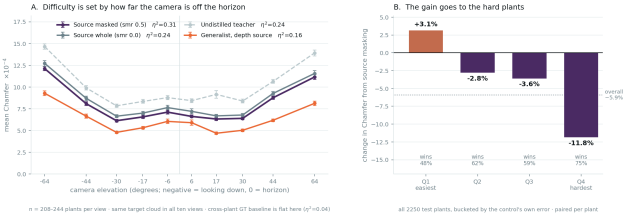



Figure 4. **Viewpoint determines the difficulty.** (A) Photograph-to-cloud Chamfer of every test plant by camera elevation (208–244 plants per view; the target cloud is the same at every view, so only the image changes). The undistilled model and three students follow the same U-shaped curve, symmetric about the horizon; elevation explains $\eta^2 = 0.24$–$0.31$ of the per-plant variance, and no shape measure of the plant correlates beyond $|\rho| = 0.06$. (B) The gain from hiding half the source's tokens during distillation, by quartile of the control's error: the gain is concentrated on the hardest plants. Add the camera-rig schematic; replace with final-release students.

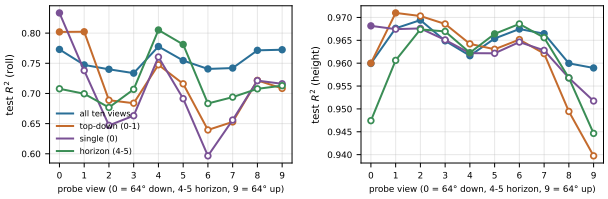



Figure 5. **Training views (E5), probe on held-out views, sorghum test.** The probe is fitted and scored on view $k$ of every plant, for the four-stream arm trained on all ten views and on three bands of the elevation ladder. Filled markers: views the arm trained on. Each restricted arm leads on roll only on its own views; height separates the arms by at most 0.02.

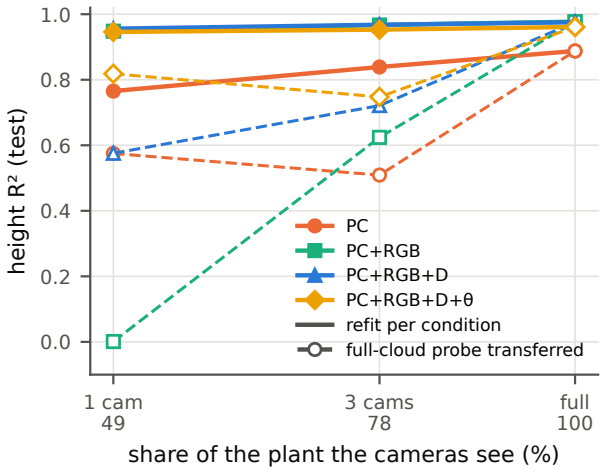



Figure 6. **The information survives partial clouds; a fixed readout does not.** Height $R^2$ (test) against the share of the plant visible to one or three cameras. Solid: probe refit per condition. Dashed: the full-cloud probe transferred as is. Sorghum, final release.

**Self-occlusion.** Training clouds are complete; a rig's are not. In Tab. 4, each model is scored on the points visible to one or three cameras, about half or three quarters of a sorghum plant. A model that sees RGB, with the probe refit per condition, loses 0.015 to 0.03 of height $R^2$ at one camera and under 0.01 at three, against −0.12 and −0.05 for PC-only. At one camera, roll loses 0.06 to 0.11 and leaf length loses 0.04 to 0.05. Depth matters more on a partial cloud than on a complete one: at one camera, PC+RGB loses −0.029 [−0.033, −0.026] on height and PC+RGB+D loses −0.020 [−0.022, −0.017]. However, the latent representation shifts with coverage (Fig. 6): a readout fitted on complete clouds and transferred to one-camera clouds collapses (PC+RGB height 0.977 → 0.624 → 0.001; headline model 0.982 → 0.623 → −0.26), and refitting recovers 0.948 and 0.960. The headline model has the best matched probe and the best complete-cloud Chamfer of all five models, but its one-camera Chamfer (11.6) is no better than the E2 arms'. The decoder also does not complete the hidden part of the plant: at one camera, the distance from the true cloud to the prediction is 11.4 to 12.8 against 4.1 for echoing the partial input, so the model reproduces what it is given rather than hallucinating the hidden half. The real-data protocol of Sec. 4.8 follows from these results: the probe will be refit on scans, and the transfer gap will

be reported as the predicted sim-to-real penalty.

**Morphological complexity.** The held-out splits are enriched with unusual morphologies by construction (Sec. 3.2), and across them neither reconstruction nor the probe degrades with plant complexity: the per-plant photograph-to-cloud error is uncorrelated with leaf count and stem length ($|\rho| \leq 0.06$ above), and the bootstrap intervals in every table are computed over these extreme-enriched plants. Fig. 4 companion: error by leaf-count decile for both species from the FIELDPROC evaluation; maize self-occlusion rows of Tab. 4.

### 4.4. Regime III: neighbor occlusion in realistic rows

The row data pairs every observation with an isolated twin (Sec. 3.2), so Tab. 5 compares each plant with itself: only the neighbors change between its isolated and row columns. The table answers three questions. How much does the zero-shot model lose when a third of its pixels and two thirds of its cloud belong to other plants (the echo test: Chamfer against the input cloud indicates whether the model reconstructs the target or the scene)? Does training against the target plant behind its neighbors recover the lost accuracy, and does the isolated control, which sees the same plants without neighbors, improve by the same amount (which would make the gain a data effect, not an occlusion effect)? And does the error follow the occlusion measure, by yaw (13 % of the target hidden broadside against 1 % edge-on) and by quartile? From Sec. 4.3 we predict that inputs including RGB lose least and that the decoder will not complete a hidden blade, so the recipe, which describes the whole plant, should be the least affected output. Fig. 7 shows held-out plants of both species in their rows.

**What the rows cost.** On maize (Tab. 5, Fig. 8), neighbors cause a large error only for a model that was not trained on rows. The isolated control, fine-tuned from the headline model, is sixteen times worse on rows than on its own clean inputs (29.0 against $1.8 \times 10^{-4}$ m$^2$, $+27.2$ $[+26.9, +27.6]$), with the leaf-recipe error higher by 0.050. The headline model without fine-tuning loses little to the neighbors ($+3.2$ $[+2.9, +3.6]$) because it is already poor under the rover camera, whose viewpoint and framing it never saw (26.4 on the isolated twins). Training against the target plant behind its neighbors removes 92 % of the row penalty: the row-trained model on rows (4.0) beats the isolated-trained model on rows by $-25.0$ $[-25.3, -24.7]$, and its error is only $+2.2$ $[+2.2, +2.3]$ above the isolated-trained model on clean inputs, with the leaf-recipe error higher by 0.008 and 0.042 lower than the isolated-trained model on rows. The first-release models gave the same picture (91 %). The row-trained model's Chamfer against the input scene rises during training while its Chamfer against the target falls, so the model reconstructs the target plant, not the row. Depth alone is the best single input for the row-trained model (9.6), the scene cloud alone is the weakest (21.9), and the error doubles from its minimum at 20 to 40° yaw (3.2) to end-on (6.4 at 90°), where the neighbors overlap the target in the image.

Sorghum repeats the result at a larger scale. The isolated-plant model is ten times worse on rows than on its own clean inputs (133.8 against $13.1 \times 10^{-4}$ m$^2$), and row training removes 85 % of that penalty: the row-trained model on rows beats the isolated-trained model on rows by $-103.1$ $[-104.3, -102.0]$, and its error is $+17.6$ $[+17.4, +17.8]$ above the isolated-trained model on clean inputs, with the leaf-recipe error higher by 0.013. Depth alone is again the best single input (18.6). Unlike maize, the sorghum error does not depend on yaw (Fig. 8d). The zero-shot headline model is worse than the isolated control even on isolated inputs (174.3 against 13.1), because the row renderer's clouds are partial captures while the headline model trained on complete clouds; this row is therefore also a partial-cloud test, as in Tab. 4.

### 4.5. Any-subset inference and distillation

The pretrained model already generates across modalities without any single-sensor training: in sorghum a photograph alone yields the cloud at Chamfer 2.27 and the recipe at MAE 0.058. In maize, the pretrained model generates well from a depth image (1.26) and poorly from a photograph alone (11.3). Distillation against the full-modality teacher then improves every source, and the gain is largest where the source was weakest (Tab. 6). The mean generation loss falls by 30.2 % in sorghum. Depth, the strongest input before distillation, improves least (Chamfer $-30$ %, recipe error $-21$ %). From a photograph, the Chamfer falls by 61 % and the recipe error by 76 %. With the cloud as the only input, the Chamfer falls by 75 % (5.48 $\rightarrow$ 1.39) and the recipe error by 50 % (0.040 $\rightarrow$ 0.020). From the recipe alone, the generated cloud's Chamfer falls by 70 %. In maize, the mean generation loss falls by 43.5 %, and the photograph row changes most (11.3 $\rightarrow$ 0.77): the weakest source again gains the most. Because the sources start so far apart, we report per-source numbers rather than only the mean. Hiding half of the source's tokens does not help the maize RGB student: the fully visible student reaches 0.76 and the two half-hidden runs 0.79 and 0.80 (val, best epoch). With every stream visible, the maize students reconstruct the cloud as well as the pretrained model (test F1 at $\tau$=0.01: 0.511 for the generalist and 0.496 to 0.502 for the RGB students, against 0.504).

A second stream helps most when it is the recipe: the students that see the recipe next to a photograph or a depth image generate the cloud at 0.40 and 0.37, against 0.88

| | | Chamfer vs. target, $10^{-4}$ m$^2$ | | | leaf-recipe MAE | | |
|---|---|---|---|---|---|---|---|
| model | input | isolated | row | Δ [95 % CI] | isolated | row | Δ [95 % CI] |
| *Maize, zero-shot (headline model, epoch 1,000)* | | | | | | | |
| | RGB | 56.7 | 59.7 | +3.0 [+2.4, +3.6] | 0.085 | 0.088 | +0.003 [+0.003, +0.003] |
| | depth | 47.2 | 62.6 | +15.3 [+14.8, +15.9] | 0.090 | 0.093 | +0.002 [+0.002, +0.003] |
| | point cloud | 20.0 | 33.8 | +13.7 [+13.4, +14.1] | 0.064 | 0.081 | +0.018 [+0.017, +0.018] |
| | all three | 26.4 | 29.6 | +3.2 [+2.9, +3.6] | 0.086 | 0.093 | +0.007 [+0.007, +0.008] |
| *Maize, row fine-tune (from the headline model)* | | | | | | | |
| | RGB | 13.7 | 14.4 | +0.6 [+0.4, +0.8] | 0.067 | 0.071 | +0.004 [+0.003, +0.004] |
| | depth | 10.3 | 9.6 | −0.7 [−0.9, −0.5] | 0.061 | 0.064 | +0.003 [+0.003, +0.004] |
| | point cloud | 21.7 | 21.9 | +0.2 [−0.1, +0.5] | 0.063 | 0.073 | +0.010 [+0.010, +0.010] |
| | all three | 12.6 | 4.0 | −8.6 [−8.8, −8.3] | 0.046 | 0.052 | +0.005 [+0.005, +0.005] |
| *Maize, isolated control (from the headline model)* | | | | | | | |
| | RGB | 11.6 | 56.1 | +44.5 [+44.0, +45.1] | 0.071 | 0.095 | +0.024 [+0.023, +0.024] |
| | depth | 4.6 | 52.8 | +48.2 [+47.5, +48.9] | 0.059 | 0.097 | +0.038 [+0.037, +0.039] |
| | point cloud | 8.2 | 25.1 | +16.9 [+16.6, +17.3] | 0.046 | 0.076 | +0.030 [+0.029, +0.030] |
| | all three | 1.8 | 29.0 | +27.2 [+26.9, +27.6] | 0.043 | 0.093 | +0.050 [+0.049, +0.051] |
| *Sorghum, zero-shot (headline model, epoch 1,000)* | | | | | | | |
| | RGB | 163.1 | 218.5 | +55.4 [+52.6, +58.4] | 0.129 | 0.135 | +0.006 [+0.006, +0.006] |
| | depth | 106.2 | 207.2 | +101.0 [+98.0, +104.2] | 0.070 | 0.109 | +0.039 [+0.039, +0.040] |
| | point cloud | 177.7 | 231.4 | +53.8 [+52.2, +55.4] | 0.114 | 0.127 | +0.013 [+0.012, +0.013] |
| | all three | 174.3 | 230.5 | +56.2 [+53.9, +58.5] | 0.108 | 0.114 | +0.006 [+0.006, +0.006] |
| *Sorghum, row fine-tune* | | | | | | | |
| | RGB | 34.8 | 29.8 | −5.0 [−5.4, −4.7] | 0.094 | 0.097 | +0.003 [+0.003, +0.004] |
| | depth | 24.4 | 18.6 | −5.8 [−6.1, −5.5] | 0.078 | 0.055 | −0.022 [−0.023, −0.022] |
| | point cloud | 46.0 | 70.8 | +24.8 [+24.4, +25.2] | 0.074 | 0.082 | +0.007 [+0.007, +0.007] |
| | all three | 38.7 | 30.7 | −8.0 [−8.5, −7.6] | 0.073 | 0.061 | −0.012 [−0.013, −0.011] |
| *Sorghum, isolated control* | | | | | | | |
| | RGB | 15.7 | 133.2 | +117.5 [+115.8, +119.3] | 0.094 | 0.114 | +0.020 [+0.020, +0.021] |
| | depth | 6.6 | 171.9 | +165.3 [+162.9, +167.8] | 0.042 | 0.100 | +0.059 [+0.058, +0.060] |
| | point cloud | 30.9 | 100.9 | +70.0 [+68.8, +71.3] | 0.065 | 0.083 | +0.018 [+0.018, +0.019] |
| | all three | 13.1 | 133.8 | +120.8 [+119.5, +122.1] | 0.049 | 0.096 | +0.047 [+0.046, +0.048] |

Table 5. **Regime III: the same plants isolated and in a row.** Every held-out plant at ten yaws under the rover camera, the target alone and the target between two bent neighbors (Sec. 3.2). Each input set alone, the recipe never an input; test split of each species, 2,250 plants × 10 yaws. Chamfer in metric units, because the row and isolated clouds are normalized to different extents. Δ = row minus isolated, paired per plant, bootstrap over plants. The row fine-tune reconstructs the target plant from the row streams; the control is the same schedule on the isolated twins. The zero-shot model of each species is its headline model, and both fine-tuned models of each species start from it.

| | Sorghum, final release (100 ep) | | | | Maize, final release (100 ep) | | | |
|---|---|---|---|---|---|---|---|---|
| | $\theta$ MAE ↓ | | Chamfer $\times 10^{-3}$ ↓ | | $\theta$ MAE ↓ | | Chamfer $\times 10^{-3}$ ↓ | |
| input | before | after | before | after | before | after | before | after |
| RGB only | 0.058 | 0.014 | 2.27 | 0.88 | 0.037 | 0.019 | 11.3 | 0.77 |
| depth only | 0.017 | **0.013** | 0.90 | 0.63 | 0.024 | 0.019 | 1.26 | 0.58 |
| PC only | 0.040 | 0.020 | 5.48 | 1.39 | 0.020 | 0.018 | 6.43 | 0.83 |
| $\theta$ only | – | – | 22.7 | 6.83 | – | – | 80.7 | 12.8 |
| *two streams → cloud (dedicated student / one student for all three pairs)*¶ | | | | | | | | |
| RGB+D | – | – | 2.91 | 0.60 / 0.59 | – | – | 1.30 | 0.54 / 0.64 |
| RGB+$\theta$ | – | – | 0.43 | 0.40 / 0.39 | – | – | 0.47 | 0.44 / 0.43 |
| D+$\theta$ | – | – | 0.50 | 0.37 / **0.36** | – | – | 0.46 | 0.42 / 0.40 |
| mean generation loss | 0.371 → 0.259 (−30.2 %) | | | | 0.290 → 0.164 (−43.5 %) | | | |

Table 6. **Any-subset results before and after distillation.** Each row: that input alone, the others absent, generating the rest (val). Before = the pretrained model used zero-shot (the student's warm start, epoch 0); after = the distilled generalist. The teacher sees all four streams, the parameters included; the student is trained on all four sources, and in each row it receives that source alone (Sec. 3.4). ¶Separate students that see both streams fully and are trained to generate the cloud only (Sec. 3.4), so they have no $\theta$ MAE; before = the same warm start with both streams visible. $\theta$ MAE on hidden real leaf tokens. Maize RGB+D (epoch 65) and three-pair (epoch 35) students still training: interim values in red.

from the photograph alone and 0.60 from the photograph with depth. One student trained on all three pairs matches the three dedicated students within 0.02 on every pair. Before distillation, the photograph with depth is worse than the photograph alone (2.91 against 2.27): pretraining never showed two streams with every token visible, and distillation removes this gap. Maize repeats the ordering: the recipe students generate at 0.44 (with the photograph) and

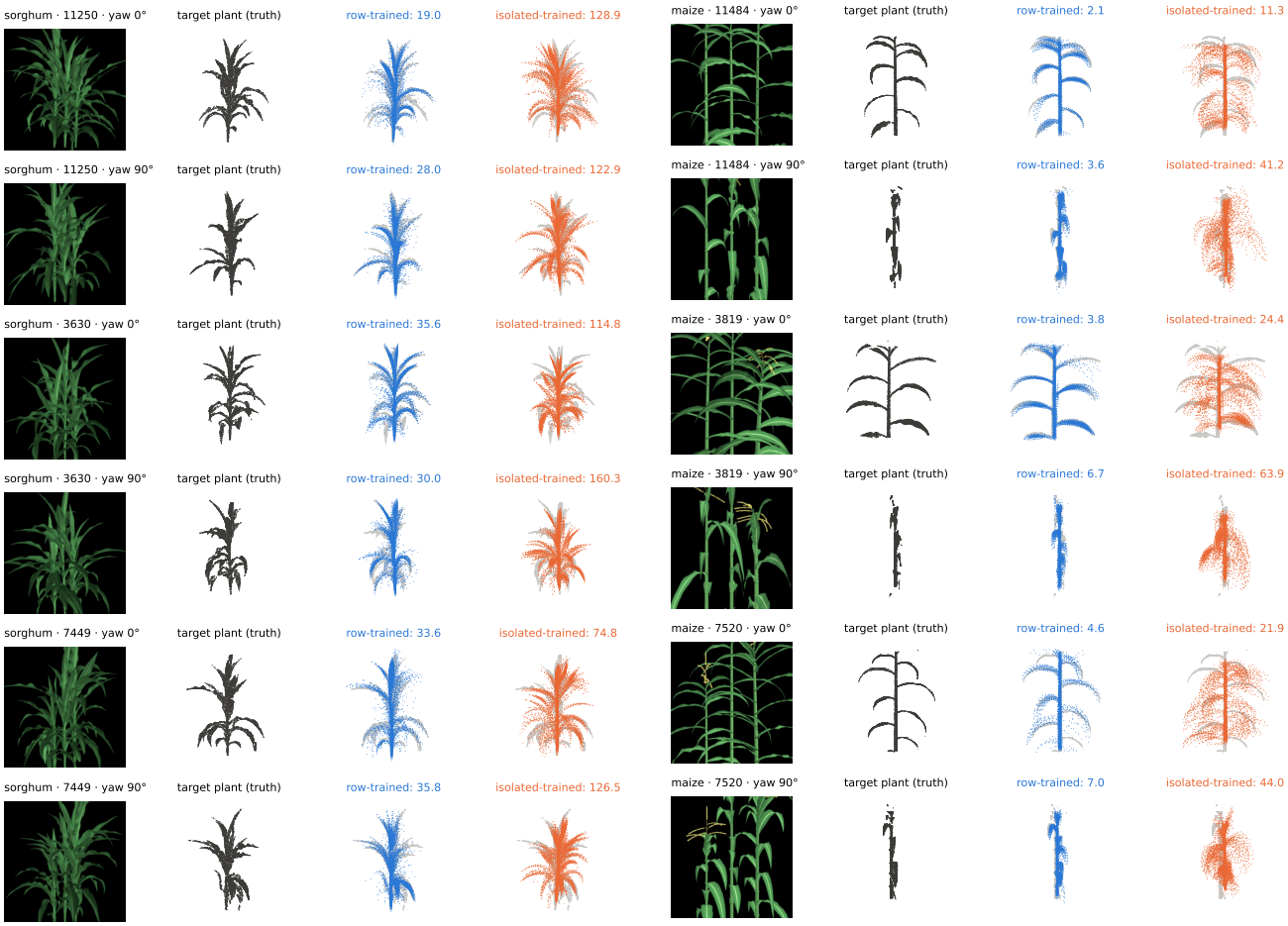



Figure 7. **Rows, held-out plants.** For each species, plants at yaw 0° (broadside) and 90° (end-on; sorghum left, maize right): the row photograph the model receives, the target plant, and the predictions of the row-trained model (blue) and the isolated-trained control (orange) over the target (gray). The control rebuilds neighboring leaves as part of the target; the row-trained model keeps to the target plant.

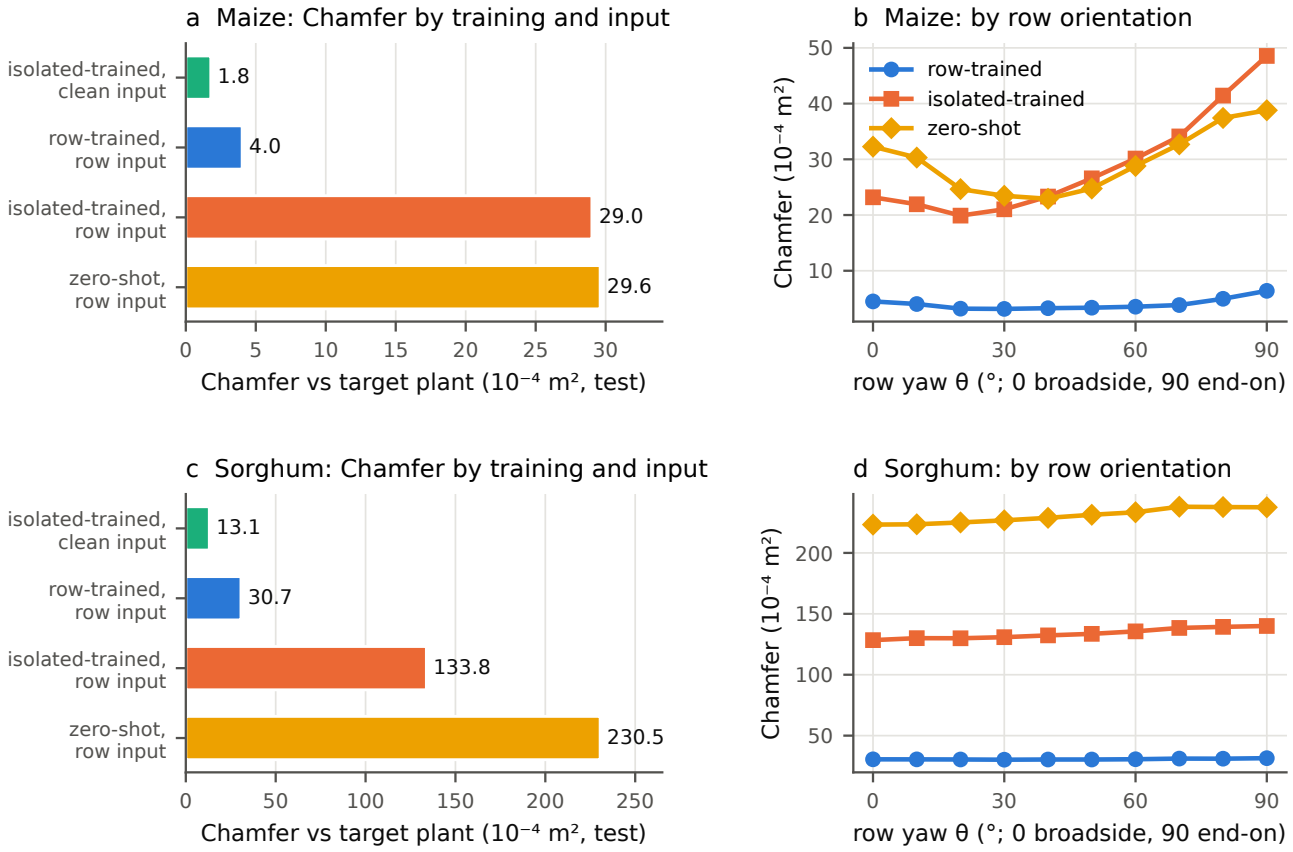



Figure 8. **Regime III on maize (a, b) and sorghum (c, d) rows, test.** (a, c) Chamfer against the target plant for the isolated-trained model on clean and on row inputs, the row-trained model on rows, and the zero-shot model on rows. (b, d) The same by row yaw. Training on rows removes most of the error added by the neighbors in both species.

0.42 (with depth), against 0.77 from the photograph alone, and the photograph with depth again starts worse than either stream alone (1.30) and improves under distillation (maize RGB+D at epoch 65: 0.54). The maize recipe students gain little from distillation (0.47 → 0.44, 0.46 → 0.42) because the pretrained model already generates well from these pairs.

On the full test split (2,250 plants, each student at its best validation checkpoint), the photograph-to-cloud Chamfer falls from 2.25 for the pretrained model to 0.88 for the RGB-only student. Hiding half of the source's own tokens during distillation beats full visibility (0.815 and 0.818 for two independent runs, against 0.88), and the fully visible student reaches its best validation point at epoch 17 of 100. The generalist, trained on all four sources, matches the dedicated student from a photograph (0.88) and generates the cloud from a depth image at 0.64. With every stream visible, the distilled models reconstruct the cloud nearly as well as the pretrained model (F1 at $\tau$=0.01: 0.375 for the generalist and 0.383 for the half-hidden students, against 0.394). On the first release, the gain from hiding half of the tokens was concentrated on hard plants: the easiest quartile lost 3 % and the hardest gained 12 % (Fig. 4B). A specialist distilled to predict the recipe from the cloud alone reached MAE 0.039 (−37 %), against 0.035 (−44 %) for the four-target generalist, so the other heads acted as auxiliary supervision for parameter regression rather than competing with it (both first release, at different learning rates, $3\times10^{-5}$ vs. $10^{-4}$; no final-release specialist was trained).

### 4.6. Where the parameter stream belongs

Table 7 and Fig. 9 compare the stream sets at equal compute. **An image stream adds most of the phenotype information.** On sorghum, adding either the photograph or the depth image to the cloud gives the largest gain in every column (height 0.895 → 0.979 with RGB and 0.895 → 0.976 with depth; roll 0.39 → 0.81 and 0.39 → 0.80), and the depth image wins all nine learnable targets over the cloud alone on both splits. On maize, the photograph gives the same jump (height 0.17 → 0.62, stem radius 0.29 → 0.75),

| | Chamfer | frozen-probe $R^2$ (val) | | | | | |
|---|---|---|---|---|---|---|---|
| streams | $\times 10^{-3}$ | height | leaves | biomass | branch | roll | leaf len |
| *Sorghum* | | | | | | | |
| PC | 5.00 | 0.895 | 0.896 | 0.892 | 0.792 | 0.393 | 0.088 |
| PC+D | 2.45 | 0.976 | 0.979 | 0.982 | 0.903 | 0.798 | 0.307 |
| PC+RGB | 2.57 | **0.979** | **0.983** | **0.986** | **0.934** | 0.808 | 0.267 |
| PC+RGB+D | **1.85** | 0.977 | 0.980 | 0.982 | 0.926 | 0.781 | **0.309** |
| PC+$\theta$ | 4.84 | 0.860 | 0.864 | 0.863 | 0.780 | 0.481 | 0.078 |
| PC+D+$\theta$ | 4.08 | 0.907 | 0.909 | 0.916 | 0.856 | 0.694 | 0.191 |
| PC+RGB+$\theta$ | 3.00 | 0.962 | 0.964 | 0.967 | 0.890 | **0.810** | 0.225 |
| PC+RGB+D+$\theta$ | 2.53 | 0.962 | 0.964 | 0.965 | 0.880 | 0.760 | 0.249 |
| | | height | leaves | biomass | angle | curl | stem r. |
| *Maize* | | | | | | | |
| PC | 2.63 | 0.171 | 0.670 | 0.610 | 0.756 | 0.114 | 0.289 |
| PC+D | 2.55 | 0.453 | 0.773 | 0.683 | 0.804 | 0.082 | 0.550 |
| PC+RGB | 1.61 | 0.620 | **0.914** | **0.783** | **0.927** | 0.334 | 0.753 |
| PC+RGB+D | 1.21 | 0.633 | **0.914** | 0.774 | 0.926 | 0.236 | 0.760 |
| PC+$\theta$ | 2.49 | 0.256 | 0.740 | 0.662 | 0.787 | 0.114 | 0.363 |
| PC+D+$\theta$ | 2.54 | 0.382 | 0.801 | 0.670 | 0.795 | 0.136 | 0.408 |
| PC+RGB+$\theta$ | 1.21 | 0.509 | 0.881 | 0.780 | 0.912 | **0.371** | **0.763** |
| PC+RGB+D+$\theta$ | **0.99** | **0.635** | 0.908 | 0.781 | 0.919 | 0.304 | 0.743 |

Table 7. **Modality value-add (E2).** Arms at equal steps, differing only in which token streams exist: every stream set that contains the cloud in both species. $\theta$ is the parameter stream as an *input* inside the shared masking budget. Probe $R^2$ on held-out plants; branch and roll are the two sorghum leaf-angle definitions. Test values are within 0.035 of val on every sorghum cell and within 0.05 on every maize cell. Both species on their final data release.

| | vision | $\theta$ | Chamfer | $R^2$ (val) | | | |
|---|---|---|---|---|---|---|---|
| arm | tokens | visible | $\times 10^{-3}$ | height | leaves | roll | leaf len |
| PC+RGB+D | 117.0 | – | 1.85 | **0.977** | **0.980** | **0.781** | **0.309** |
| +$\theta$, shared budget | 107.4 | 58 % | 2.53 | 0.962 | 0.964 | 0.760 | 0.249 |
| +$\theta$, gated 20 % | 117.0 | 20 % | 2.29 | 0.964 | 0.967 | 0.771 | 0.278 |
| +$\theta$, gated 60 % | 117.0 | 60 % | 2.40 | 0.959 | 0.963 | 0.761 | 0.239 |
| +$\theta$, target only | 117.0 | 0 % | 2.49 | 0.966 | 0.970 | 0.702 | 0.220 |
| *Maize* | | | | height | leaves | curl | stem r. |
| PC+RGB+D | | – | 1.21 | 0.633 | 0.914 | 0.236 | 0.760 |
| +$\theta$, shared budget | | | 0.99 | 0.635 | 0.908 | 0.304 | 0.743 |
| +$\theta$, gated 20 % | | 20 % | | 0.704 | 0.931 | **0.464** | 0.791 |
| +$\theta$, gated 60 % | | 60 % | | 0.521 | 0.851 | 0.303 | 0.688 |
| +$\theta$, target only | | 0 % | | **0.776** | **0.947** | 0.357 | **0.804** |

Table 8. **Token-budget control.** The four-stream arm re-trained with the parameter tokens outside the shared Dirichlet budget, so the three sensor streams see the same visible-token count as the three-stream arm (117.0; seed-matched masks identical in 5,000 of 5,000 draws), at three parameter visibilities. "Target only" never encodes a parameter token. The target-only and 60 % arms lose all nine learnable targets to PC+RGB+D on both splits, and the 20 % arm loses eight (maximum leaf length is a draw). In maize the 20 % and target-only arms beat both the shared-budget arm and PC+RGB+D on all eleven traits on both splits, and the 60 % arm loses nine. Intervals are in the supplement. Fig. 10 plots the sorghum rows.

and the depth image a smaller one (height $0.17 \rightarrow 0.45$; nine of eleven traits won, curl lost). The depth image records the same surface as the cloud, so its gain does not come from new information about the plant. A likely reason is that dense, ordered image patches present the geometry in a form that the encoder reads more easily than sampled point tokens. Once depth is present, the photograph adds little: paired over plants, PC+RGB+D wins leaf width ($+0.07$, test) and leaf count (by 0.001) against PC+D, loses roll ($-0.015$) and maximum leaf length ($-0.06$), and ties the other five targets. **Added to RGB, depth improves reconstruction but not the phenotype readout:** depth lowers the Chamfer by 28 % in sorghum and by 25 % in maize. On sorghum, it changes the probe by thousandths: paired over plants, it loses on size ($-0.002$ to $-0.004$) and on the two angles, and gains only on leaf length ($+0.05$). On maize (final release, paired over plants), depth wins no trait and loses three on both splits, namely tassel droop ($-0.10$ to $-0.12$), curl ($-0.06$ to $-0.10$) and twist ($-0.03$), and the other eight traits tie within 0.015. **The parameter stream, used as an input, does not help.** On sorghum the four-stream arm loses to PC+RGB+D on all nine learnable targets on both splits (size $-0.015$ to $-0.018$, branch $-0.038$, roll $-0.017$, leaf length $-0.075$, leaf width $-0.14$; every interval excludes zero), and its Chamfer, 2.53 against 1.85, loses the improvement that depth brought. The other three sorghum stream sets show the same pattern. Added to PC+D, the parameter stream loses all nine targets; added to PC+RGB, it loses seven and ties roll and maximum leaf length; added to the cloud alone, it wins roll ($+0.10$) and loses the three size traits ($-0.03$). It raises the Chamfer in every set that contains an image stream ($2.45 \rightarrow 4.08$ with depth, $2.57 \rightarrow 3.00$ with RGB) and lowers it only for the cloud alone ($5.00 \rightarrow 4.84$). On the first maize release, the same arm lost 10 of 11 traits. On the final release, the paired comparison is a draw: PC+RGB+D wins droop ($+0.04$ to $+0.07$), loses curl ($-0.04$ to $-0.07$) and twist ($-0.035$), and ties the eight remaining traits within 0.02 (height 0.635 vs. 0.633, stem radius 0.743 vs. 0.760), while the four-stream arm has the better Chamfer. Against PC+RGB, the four-stream arm loses droop ($-0.05$) and tassel droop ($-0.13$) and ties the rest. The new maize arms do not support a uniform rule either: added to the cloud alone, the parameter stream wins seven traits and loses none, added to PC+RGB it wins three (twist, curl, tassel droop) and loses five, and added to PC+D it wins three and loses two. Either way, the stream that encodes the answer verbatim in training is not the stream that makes the frozen feature better at deployment, where it is absent.

A fourth stream at mask ratio 0.8 splits the visible budget four ways, so the comparison in Tab. 7 alone cannot separate a token-budget handicap from a representational one: as trained, the four-stream arm's sensors see 107.4 visible tokens against 117.0, and its parameter tokens are 58 % visible. In the control arms of Tab. 8, the handicap is removed. Restoring the sensor budget does not recover the three-stream arm's probe scores at any parameter visibility; the trend with visibility is monotone on the size targets

(60 % < 20 % < 0 % visible), and making the parameters a pure reconstruction target still loses 0.011 on size, 0.07 on roll, 0.10 on leaf length and 0.15 on leaf width. The per-leaf traits stored in the parameter tokens (roll, leaf length and leaf width) lose the most. The pattern is consistent with the encoder routing these traits through tokens that do not exist in the field, and the target-only result shows that the reconstruction pressure alone accounts for part of the loss.

Maize gives the opposite result for the gated arms (Tab. 8). With the sensor budget restored, the arm that sees 20 % of the parameter tokens and the arm that only reconstructs them beat both the shared-budget arm and PC+RGB+D on all eleven traits on both splits (target only, val: height 0.776 against 0.633, stem radius 0.804 against 0.760), while the arm that sees 60 % loses nine traits. Making most of the parameter stream visible to the encoder therefore hurts in both species, but a parameter target hurts the sorghum representation and helps the maize one. The headline models were nevertheless trained with the parameters in the encoder's shared budget (Sec. 3.3), as in the four-stream arm of Tab. 7. For sorghum, the ablations predict that the headline model's frozen representation is weaker than that of a three-stream or $\rho_\theta$=1 model; for maize, they predict the opposite of $\rho_\theta$=1. We have not tested either at the headline scale. Supervising the parameters as a target, in turn, lets the generalist beat the parameter specialist (Sec. 4.5).

### 4.7. What the metrics disagree about

Chamfer and the probe rank the arms alike only in the coarse sense (Fig. 11; Spearman −0.57 to −0.88 across twelve arms), and they disagree on every design decision in this paper. Depth is the clearest case: −28 % Chamfer in sorghum, −25 % in maize, and a probe that does not move (Sec. 4.6). Data is the second: sorghum Chamfer keeps improving from 1,000 to 10,000 plants (−21 %) while the size targets stay within 0.93–0.96 and are not monotone (Tab. 10), and in maize Chamfer falls fourfold while the probe is not even monotone. The parameter stream is the third: on the final maize release it gives the best Chamfer of the four arms and no probe gain. And the partial clouds of Tab. 4 are the fourth, in the other direction: at one camera Chamfer is 2.6 to 6.5× worse while the refit probe loses at most 0.03 on height. Point-MAE, an external baseline, beats our PC-only arm on every sorghum trait under the probe (Tab. 9), so our advantage comes from fusing a photograph with geometry, not from a better cloud encoder.

A nearest-neighbor distance is minimized by surface proximity and allows many predicted points to match one target point, so it rewards a decoder that places points on the bulk of a slender plant and barely penalizes a missing or merged blade [43, 48]. The probe instead measures the structure retained by the encoder, which is the input to a trait pipeline. We therefore use the probe to decide between modality and data choices and report Chamfer as the decoder's surface error. On the training side, we add the Sinkhorn term of Sec. 3.3, which matches the two clouds as distributions and is not minimized by a clump of points. Add the E7 ablation of that term, and F-score at 1 cm beside Chamfer in Tabs. 3 and 6 (student F-scores queued).

**F-score.** Precision and recall at a distance threshold expose the failure hidden by Chamfer (Fig. 12). For every model, precision is far higher than recall: the headline model places 51 % of its points within 1 cm of the plant but covers only 32 % of the plant's points (F1 0.39; 0.72 at 2 cm). The reconstruction is accurate where it places points but misses whole regions, namely the thin blades that a nearest-neighbor distance barely penalizes. The ablation arms rank the same under F1 as under Chamfer, so F1 adds an interpretable unit but does not change any decision.

### 4.8. Real-world validation

E10, not yet run. Protocol of Sec. 3.5: the single-sensor students on field-grown maize and sorghum whose procedural ground truth was reconstructed by the staged scan pipeline [18, 25], including row plots. Table: per input (photograph, depth frame, one-scanner cloud), recipe error against the pipeline's parameters, Chamfer / F-score of the re-rendered plant against the scan, and the pipeline's own ∼10 min per plant against one forward pass. Fig. 7: a real scan, the recovered recipe re-rendered, and the pipeline's fit. The one-camera transfer gap of Tab. 4 is the predicted penalty; the probe is refit on the scans as Sec. 4.3 shows it must be.

### 4.9. Representation quality and parsimony

**Against published encoders.** In Tab. 9, four public checkpoints and a supervised network are read through the same probe. On sorghum the three-stream CROPCODE beats every published encoder on every trait in Tab. 9; against the strongest, EmbodiedMAE-B, it wins eight of nine paired over plants: +0.004 to +0.007 on size (tight intervals at a 0.98 ceiling), +0.03 on branch, +0.04 on roll, +0.08 on leaf length and +0.09 on leaf width. The four-stream arm loses the size traits to EmbodiedMAE-B by 0.008 to 0.011: the parameter-stream penalty of Sec. 4.6 measured against an external model. A network of our architecture trained from scratch on the probe targets reaches 0.974 height and 0.767 roll through its own head; the frozen three-stream feature beats it on seven of nine traits and loses none (paired, CLS token; under mean pooling the supervised network wins branch), while the frozen four-stream feature loses size and branch. Pretraining helps without the parameter stream; with it, the frozen feature is slightly worse than training from scratch. On the final

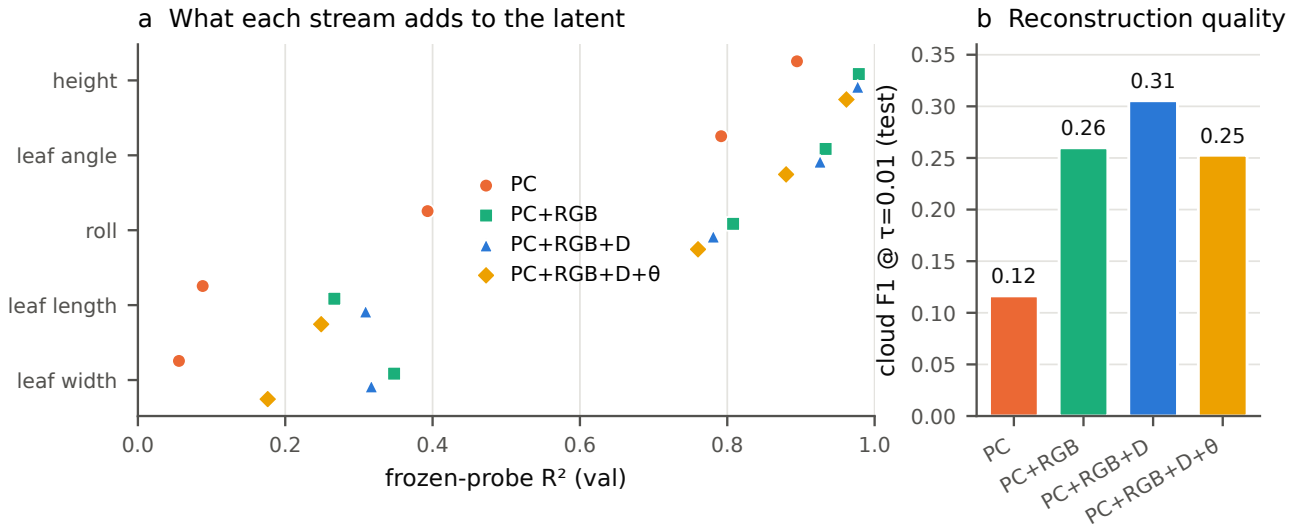



Figure 9. **What each stream adds (sorghum, final release).** (a) Frozen-probe $R^2$ (val) of the four stream arms on five traits; size traits saturate once RGB is present; the leaf traits separate the arms. (b) Reconstruction F1 at $\tau$=0.01 (about 1 cm), test: depth improves reconstruction, and the parameter stream reverses that gain.

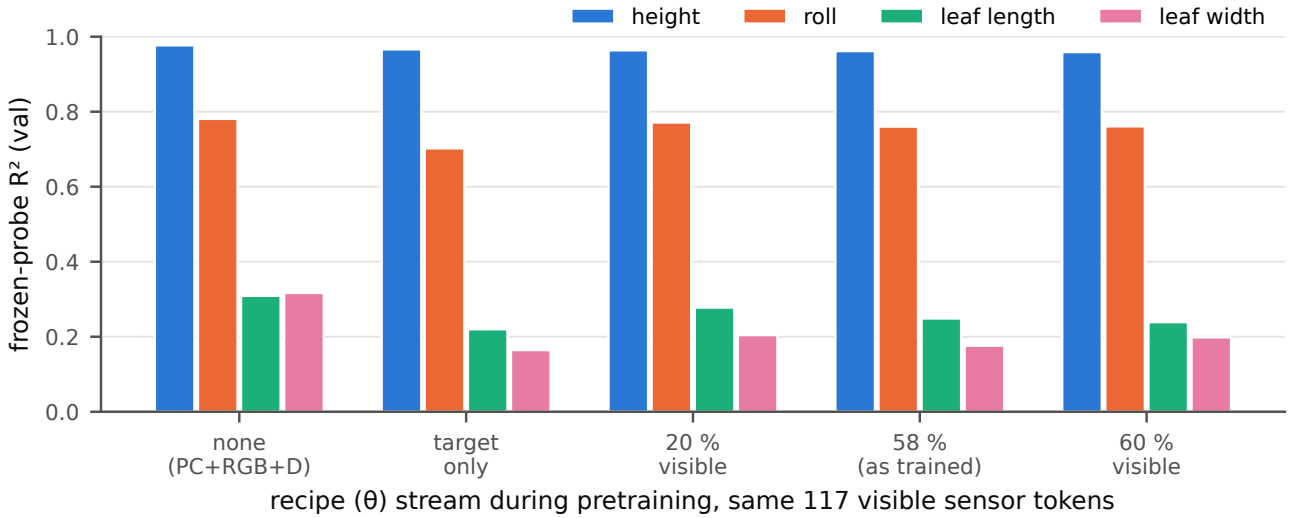



Figure 10. **Where the parameter stream belongs.** Probe $R^2$ (val) with the recipe stream absent, as a reconstruction target only, or 20 and 60 % visible during pretraining at an equal 117 visible sensor tokens, against the arm as trained (58 % visible, 107.4 sensor tokens). No variant reaches the three-stream arm on any trait (Tab. 8).

maize release (val, point estimates), the four-stream arm beats EmbodiedMAE-B on 10 of 11 traits and the best baseline per trait without camera pose on 9, losing leaf count to Point-MAE (0.908 against 0.926) and tassel droop to DINOv2 ($-0.048$ against 0.297). The maize headline model beats every such baseline on 10 of 11 (height 0.739 against 0.570, curl 0.698 against 0.199, stem radius 0.830 against 0.588) and still loses tassel droop to DINOv2 (0.215 against 0.297), the trait where every encoder of the EmbodiedMAE design is weakest. Unlike sorghum, the maize network trained from scratch on the probe targets is ahead of every frozen 600-epoch arm on all eleven traits (mean pooled: height 0.785, curl 0.618), and the maize headline model closes most of that gap and passes it on curl (height 0.739, curl 0.698 on the CLS token).

| | Sorghum | | | | | Maize | | | |
|---|---|---|---|---|---|---|---|---|---|
| encoder (frozen, mean-pooled) | height | leaves | branch | roll | leaf len | height | angle | curl | stem r. |
| DINOv2 ViT-B/14 (RGB) [33] | 0.924 | 0.928 | 0.840 | 0.573 | 0.235 | 0.561 | 0.816 | 0.072 | 0.588 |
| MultiMAE-B (RGB+D) [3] | 0.922 | 0.925 | 0.833 | 0.679 | 0.232 | 0.570 | 0.783 | 0.069 | 0.442 |
| Point-MAE (PC) [34] | 0.958 | 0.963 | 0.863 | 0.716 | 0.188 | 0.509 | 0.863 | 0.199 | 0.426 |
| EmbodiedMAE-B (RGB+D+PC) [10] | 0.976 | 0.979 | 0.871 | 0.765 | 0.265 | 0.528 | 0.823 | 0.139 | 0.428 |
| supervised from scratch (ours, 4 streams) | 0.973 | 0.978 | **0.924** | 0.634 | 0.271 | **0.785** | **0.935** | **0.618** | **0.860** |
| CROPCODE PC only | 0.906 | 0.909 | 0.801 | 0.444 | 0.107 | 0.171 | 0.756 | 0.114 | 0.289 |
| CROPCODE PC+RGB+D | **0.979** | **0.983** | 0.903 | **0.809** | **0.337** | 0.633 | 0.926 | 0.236 | 0.760 |
| CROPCODE four streams | 0.965 | 0.968 | 0.888 | 0.779 | 0.288 | 0.635 | 0.919 | 0.304 | 0.743 |

Table 9. **Frozen-encoder baselines (E8).** Published checkpoints are evaluated on the same held-out plants with the same ridge probe; mean pooling for every model, the only pooling available for all of them. For the supervised row, our architecture is trained end to end on the probe targets and evaluated with the same frozen probe. Both species on the final release. The maize rows of our three arms use the CLS token (mean-pooled maize rows not run); the maize supervised row is mean-pooled, as in sorghum.

| | | Sorghum | | | | Maize | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | Chamfer | height | roll | leaf len | Chamfer | height | angle | stem r. |
| *Training plants (E3), base model* | | | | | | | | | |
| 1,000 | 6,169 ep | 3.13 | 0.952 | 0.716 | 0.249 | 3.54 | 0.393 | 0.851 | 0.593 |
| 3,000 | 2,100 ep | 2.86 | 0.932 | 0.679 | 0.213 | 1.46 | 0.524 | 0.867 | 0.624 |
| 10,000 | 631 ep | 2.48 | 0.962 | 0.749 | 0.245 | 0.87 | 0.427 | 0.853 | 0.717 |
| 10,500 | 600 ep | 2.53 | 0.962 | 0.760 | 0.249 | 0.99 | 0.635 | 0.919 | 0.743 |
| *Model size (E4), all plants* | | | | | | | | | |
| small | 25 M | 2.86 | 0.964 | 0.817 | 0.277 | 1.22 | 0.563 | 0.886 | 0.682 |
| base | 114 M | 2.53 | 0.962 | 0.760 | 0.249 | 0.99 | 0.635 | 0.919 | 0.743 |
| large | 332 M | 2.68 | 0.948 | 0.711 | 0.246 | 1.06 | 0.580 | 0.887 | 0.614 |

Table 10. **Data and model scaling at equal steps.** Every row is 197,400 steps; epochs differ so that steps do not. Chamfer $\times 10^{-3}$ and probe $R^2$ (val) of the four-stream arm. The 10,000- and 10,500-plant rows differ by 5 % in data and are the run-to-run error bar. All rows are on the final releases; the base row of each species is its E3 full-data row. Fig. 13 plots the sorghum rows.

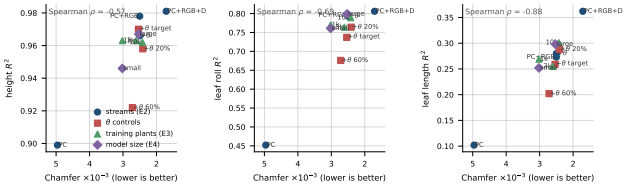



Figure 11. **Chamfer against the frozen probe, twelve sorghum arms** (streams, parameter controls, training plants, model sizes; val). The metrics agree on the gross order and disagree on the decisions: depth (PC+RGB → PC+RGB+D) moves Chamfer −32 % and height $R^2$ by +0.003; more plants (1k → 10k) moves Chamfer −20 % and does not change the probe; the parameter stream brings the Chamfer back to that of PC+RGB, with a worse probe.

**Camera pose, a forensic note.** Point-MAE given each view's full camera pose beats our camera-frame four-stream arm on 8 of 11 maize traits (height 0.720 vs. 0.597). Most of this advantage is a generator artifact: the maize renderer puts every plant's leaf plane in the same world plane, so a full pose rotation gives the encoder a canonical plant azimuth that no real rig provides. With gravity only (the camera leveled and its azimuth kept, the information a calibrated rig provides), Point-MAE's height falls to 0.588 and camera-frame PC+RGB+D wins five traits (height, biomass, twist, width, stem radius +0.23) and loses four (leaf angle, leaf count, curl −0.18, tassel droop −0.21). Our fused arms do not improve with leveled clouds, and the PC-only arm improves on five traits. A generator's unintended regularities are a benchmark hazard, and this one was found only because the baseline scored suspiciously well.

**Data, capacity, and the noise floor.** Two near-duplicate runs, 10,000 and 10,500 training plants, differ by 2 % in Chamfer and at most 0.011 in probe $R^2$; we take this difference as the noise floor. Relative to this floor (Tab. 10, Fig. 13), sorghum reconstruction improves steadily with data (3.13 → 2.86 → 2.48), while the probe is flat and not even monotone: height is 0.952/0.932/0.962, and the dip at 3,000 plants exceeds the floor. Paired over plants against the full-data arm, the 10,000-plant arm wins one trait and loses none, while the 3,000- and 1,000-plant arms lose six and four (test height −0.029 and −0.008). Model size reverses the usual trend on the probe: paired over plants, the small encoder beats base on five traits and loses none (roll +0.050, leaf length +0.040, leaf width +0.047), and the large encoder loses five (size −0.015, roll −0.046), while base has the lowest Chamfer (2.86/2.53/2.68). At this step budget a bigger encoder reconstructs no better and encodes the leaf traits worse.

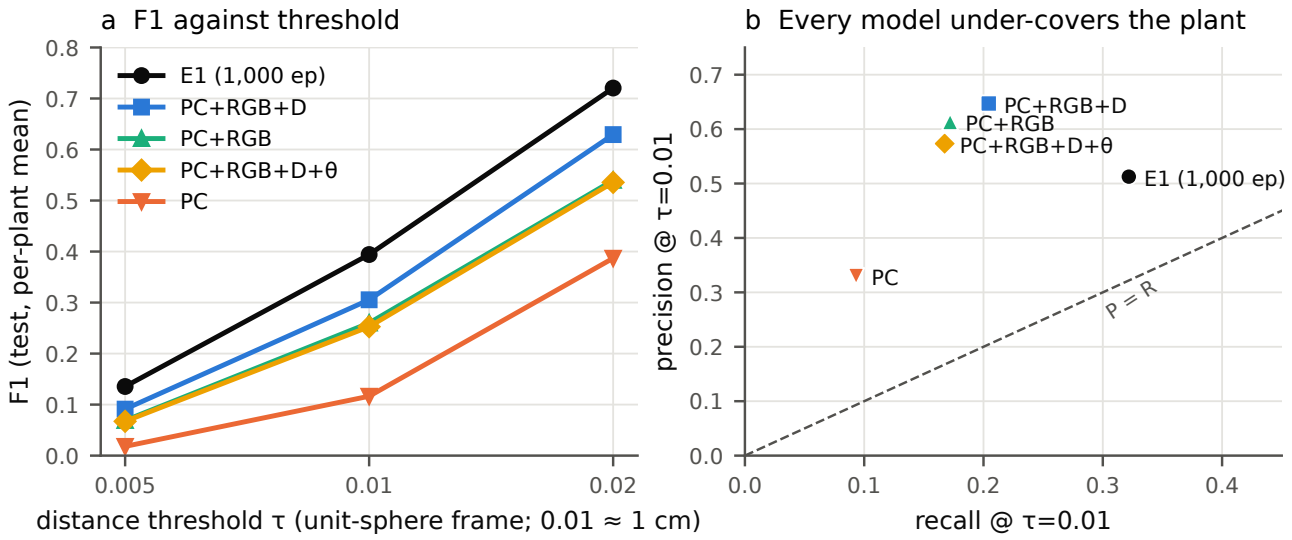



Figure 12. **Point-cloud precision, recall and F1 (sorghum, test, 2,250 plants).** (a) F1 against the distance threshold $\tau$ in the unit-sphere frame ($0.01 \approx 1$ cm). (b) Precision against recall at $\tau$=0.01: every model lies far above $P$=$R$, i.e. under-covers the plant. Reconstructions in each model's training regime (mask ratio 0.8).

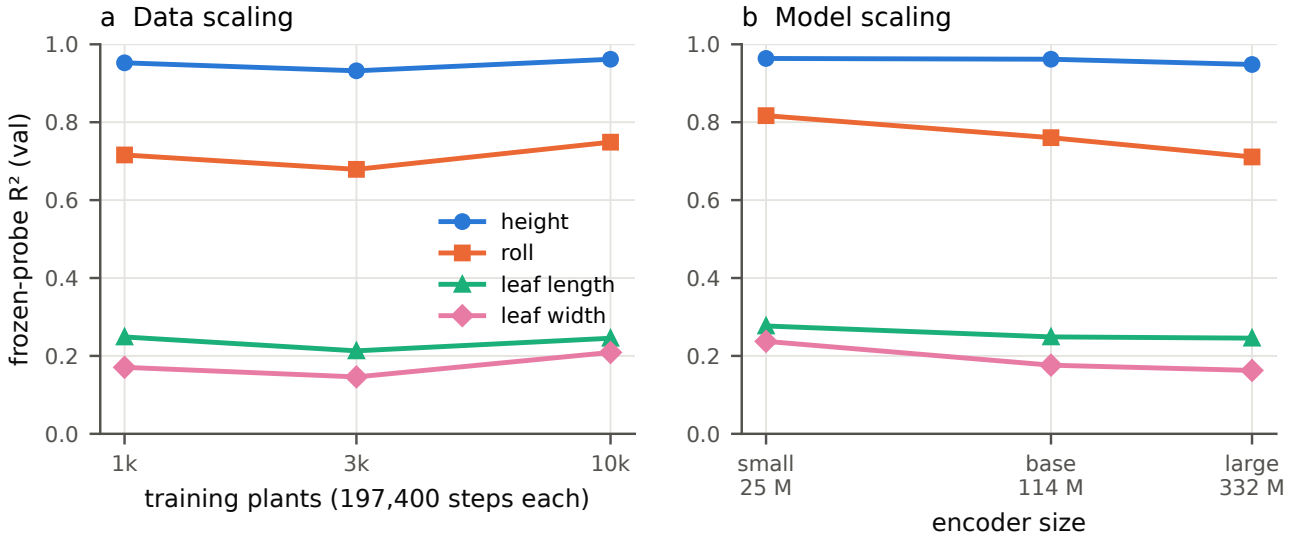



Figure 13. **Data and model scaling at equal steps (sorghum, final release).** Probe $R^2$ (val) against training plants (a) and encoder size (b), every point 197,400 steps. Neither more plants nor more parameters improves the probe at this budget.

Maize is noisier: Chamfer falls fourfold with data but the probe is not monotone (height 0.39/0.52/0.43/0.64). Paired over plants, the small and the large maize encoders each lose ten of eleven traits to base, so base is the best size in maize as well. According to the paired bootstrap, the dip is not sampling noise: every reduced arm loses to the full-data arm on 10 of 11 traits on both splits (twist ties), by 0.03 to 0.24, and the 3,000-plant arm beats the 10,000-plant arm on height (+0.10), leaf length (+0.10) and leaf count (+0.04) while losing stem radius (−0.10), each interval excluding zero on val and test. With one seed per arm, the remaining explanation is run-to-run variance of the arm itself.

A second seed of the maize 10k arm. The maize headline model, trained for 1,000 epochs on all ten views at global batch 256 with the Sinkhorn term, scores far above every 197,400-step maize arm (val: height 0.739 against 0.635 for

the four-stream full-data arm, curl 0.698 against 0.304, leaf angle 0.958 against 0.919, stem radius 0.830 against 0.743; reconstruction F1 at $\tau$=0.01, test, 0.504 against 0.314). In sorghum, against its four-stream full-data arm, the same change adds 0.021 on height (0.983 against 0.962). The comparison mixes schedule, batch and loss, so it does not isolate any one of them.

**Across species.** We pretrained the three-stream model (PC+RGB+D) on each species for 150 epochs, either from scratch or from the other species' pretrained weights (sorghum from the first-release maize model at epoch 600, maize from the sorghum headline model). Paired over plants, the warm start wins every learnable trait on both splits in both directions, nine of nine in sorghum and eleven of eleven in maize. The largest gains fall on the traits that the scratch run barely learns (test: sorghum roll +0.18 and leaf width +0.16; maize curl +0.49, stem radius +0.32 and leaf width +0.25). The warm start also lowers the validation Chamfer, from 3.38 to 2.29 $\times$ $10^{-3}$ in sorghum and from 2.55 to 0.85 $\times$ $10^{-3}$ in maize. Row reconstruction does not transfer: on the other species' test rows, a row-trained model reaches about nine times the Chamfer of the native row model (268 against 30.7 $\times$ $10^{-4}$ m$^2$ on sorghum rows, 61.7 against 6.5 on maize rows).

The maize headline model separates the two halves of the model on sorghum plants. Its frozen encoder, probed on sorghum without fine-tuning, reaches the native sorghum models on size (val height 0.976, against 0.977 for the sorghum three-stream arm and 0.983 for the sorghum headline model) and leaf width (0.360, against 0.301 and 0.415), and loses 0.02 to 0.09 on roll and leaf length. Hiding the depth image changes no trait by more than 0.016, and so does serving the sorghum photographs without their brightness gain, so the result does not depend on depth scale or exposure. Its decoder does not transfer: from a sorghum photograph it generates a cloud at 18.6 $\times$ $10^{-3}$ against 2.25 for the sorghum headline model, its reconstruction from the sorghum clouds stays near 37 $\times$ $10^{-3}$ whatever part of the plant the cameras see, and on sorghum rows it reaches 198 (zero-shot) and 237 (the maize row model) against 30.7 $\times$ $10^{-4}$ m$^2$ for the sorghum row model. The encoder therefore learns plant structure that carries across species, while the decoder learns one species' shape. T1 warm start from the maize headline model (150 epochs) training.

**What the latent encodes.** There is no genotype in either generator: every plant is one draw from one continuous distribution, so a clustered latent would be a defect, not a finding. The frozen CLS space is a continuum (best silhouette 0.15 to 0.24 at $k$=2, adjusted Rand index $\approx$ 0 against shape deciles, t-SNE trustworthiness 0.95 to 0.99) of low effective dimension: participation ratio is 4.4 to 6.3 (of 768) in sorghum and 6.7 to 11.3 in maize. Once RGB is present, the number of principal components needed for 90 % of the variance is 14 to 15 in sorghum and 18 to 19 in maize, consistent with maize's more independent generative factors. In sorghum, the leading latent direction is the plant's size (first component vs. leaf count $|r|$ = 0.73 to 0.94 across the arms); in maize, no single component aligns with a trait (best single-component correlation with leaf count $|r| \leq 0.17$ in every arm, although the probe reads leaf count at $R^2$ = 0.91 from the same features), so the maize traits are spread over many directions. The parameter stream changes the sorghum latent geometry in the way the probe results predict: the gated arms are the least aligned with size ($|r|$ 0.48 and 0.34) and the most isotropic (participation ratio 8.5 and 14.3), and the target-only arm is the most dominated by size of the five (first component 43 % of variance, $|r|$ = 0.95). Linear decodability is also distinct from local embedding structure: the probe reads roll at $R^2$ 0.81 from a latent in which roll is barely organized locally, so we use the probe, not t-SNE smoothness, as the measure for representation claims on a continuous generator (Fig. 14).

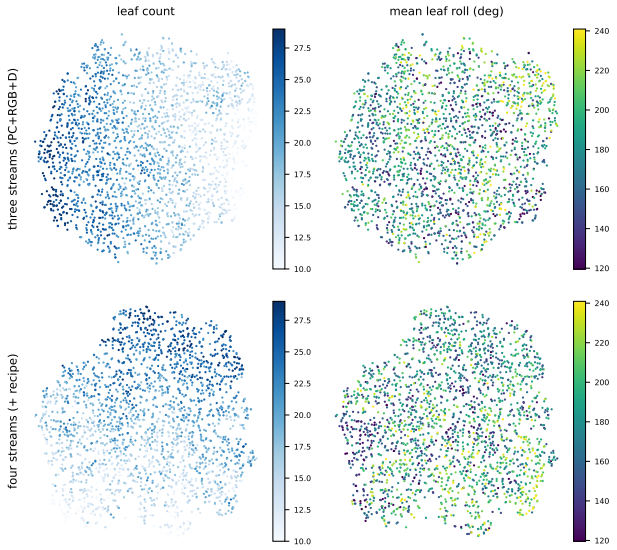



Figure 14. **Latent structure (sorghum, val).** t-SNE of the frozen CLS token of the three-stream and four-stream arms, colored by leaf count (the size axis) and by mean leaf roll (the axis independent of size). Both latents are continua; leaf count varies smoothly across each map, while roll, which the probe reads at $R^2$ 0.78 from the three-stream latent, shows no local organization.

## 5. Discussion and Limitations

**Why a stream that holds the answer hurts.** The parameter stream is the only input that states the phenotype verbatim, and it is the one input never available in the field. In

sorghum, every arm in which the encoder saw the parameter stream learned a worse sensor representation, and on the size traits the loss grew with the stream's visibility (Tab. 8). Even as a pure decoder target, the parameter stream lowered roll, leaf length and leaf width, the traits stored per leaf in the parameter tokens. Maize agrees only at high visibility: its arm that sees 60 % of the parameter tokens loses nine traits, but its arms at 20 % and as a target only improve all eleven. One explanation, consistent with modality competition and negative transfer, is that a privileged stream lets the encoder route a trait through tokens that are absent at test time, and the reconstruction pressure of a per-leaf target then shifts the latent toward the plant's size, which is already the direction of its first principal component. The ablations of both species support a narrower rule: a generative modality should not be largely visible to the encoder, and whether it helps as a target depends on the generator. Our sorghum headline model does not yet follow this rule: it was pretrained with the parameters as an encoder input, and its distillation used the same model as teacher, so its frozen representation corresponds to the four-stream arm. A headline run at $\rho_\theta{=}1$ would be the direct test. We expect the rule to transfer to any multimodal model with access to a program, a CAD sketch, or a simulator state, and the gated ratio $\rho_\theta$ is a one-line way to test it.

**Which metric decides.** Chamfer and the frozen probe agree on the coarse ranking of arms and disagree on the design decisions (Fig. 11): depth added to RGB, more training plants, and the parameter stream each move Chamfer by tens of percent and leave the probe nearly unchanged, while a partial cloud from one camera raises Chamfer by 2.6 to 6.5 times but changes the probe by at most 0.03. On slender geometry, Chamfer measures surface proximity and the probe measures recovered structure, so we use the probe to make decisions and report Chamfer as the decoder's surface error. The probe has its own limitations: it requires labels and a held-out split, and its $R^2$ is comparable between models on one split but is not an absolute measure across datasets (our held-out plants are extreme-enriched). Sorghum's four named traits also reduce to two generator variables, so our claims about independent traits rely on maize. On the training side, we added the Sinkhorn term because Chamfer barely penalizes a missing blade; its ablation is pending (E7).

**From complete clouds to a camera.** Every sorghum model was pretrained on complete clouds of isolated plants, and Sec. 4.3 measures the effect of partial clouds. Maize is the other case: a maize cloud is the surface visible to its own camera (2 % of a cloud's points hidden from that camera, 25 % from another; the depth image back-projects to within 1 mm), so every maize model trained on single-camera clouds. Fusing the clouds of three or ten cameras raises coverage of the ten-camera cloud from 72 % to 96 % and 100 %, but it moves the maize headline model off its training distribution: its reconstruction of the ten-camera cloud worsens from 1.15 to 4.54 and $4.98 \times 10^{-3}$, and its refitted height probe from 0.743 to 0.670 and 0.682 (val). A model is best on the coverage it was trained on, in both directions. The phenotype information survives a single camera's view, but the latent shifts with coverage, so a readout calibrated on complete scans is wrong on partial ones and must be refit. The decoder reproduces what it is given and does not hallucinate the hidden half. Viewpoint, not morphology, determines the difficulty of inverting a photograph. The recipe tolerates partial views best: the recovered insertion heights of one plant vary by a tenth of their spread across its ten photographs, and a re-rendered recipe is complete even where the cloud is not. In both species, training on rows removes most of the neighbor penalty (92 % in maize and 85 % in sorghum, Sec. 4.4). The real scans (Sec. 4.8) are not yet evaluated; until they are, the one-camera transfer gap is our prediction for the sim-to-real penalty. A generator also contains unintended regularities, such as the fixed leaf azimuth that gave a pose-aware baseline a canonical orientation, and benchmark users should expect more of them.

**What distillation does and does not buy.** Distillation from a full-modality teacher improves every single-sensor student and concentrates the gain on the weakest sources and the hardest plants (Tab. 6, Fig. 4B), and supervising all four targets beats a parameter specialist at parameter regression. Distillation does not make the model view-robust: the elevation curve moves down and keeps its shape, so view robustness requires a different method. The training-view ablation points to the pretraining draw: the arm trained on all ten views is the most even across elevations, and every arm trained on a narrower band loses roll outside it (Fig. 5). Distillation also showed that an unpredictable field is not neutral: the three waviness parameters of the first sorghum release stayed at zero skill and became more view-dependent under distillation, and the final generator does not include them. A parameterization meant to be regressed should be restricted to the fields that the sensors can determine.

**Scope and compute.** The ablations use one seed per arm. The run-to-run floor comes from a single near-duplicate pair (2 % in Chamfer, at most 0.011 in $R^2$), and the maize scaling differences, although significant under the paired bootstrap, could still be run-to-run variance (Sec. 4.9). Two generators and two species-specific models are studied, and the 1,000-epoch headline model is trained once per species. The procedural prior that makes inversion tractable also limits it: a plant that the generator cannot express cannot

be recovered, and an organ class that the generator renders without variation (tassel droop in maize is below zero for every encoder of this design) limits the data as much as the model. Real scans and the FIELDPROC release at 10K–100K plants remain in progress.

## 6. Conclusion

CROPCODE recovers a complete, editable procedural plant from the data of whichever sensor a field campaign used, in one forward pass, by using the generator for both the training data and the output representation. Two findings from its ablations apply beyond plants: the generator's parameters harm the representation when most of them are visible to the encoder, and on slender geometry a downstream probe, not Chamfer distance, should decide what a modality is worth. A single camera's partial view requires refitting the readout and leaves a measured transfer gap; the row-occluded and real-scan evaluations will show whether the generator's prior covers the remaining difference to field data. Sharpen once E10 lands.

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# Procedural Reconstruction of Slender Plants from Any Field Modality

## Supplementary Material

## A. Detailed Literature Review

This appendix expands the compressed related-work discussion of Section 2 to the six bodies of work on which the paper builds or from which it departs.

### A.1. Procedural plant models and inverse procedural modeling

Procedural modeling of plants began with L-systems [36] and matured into generators whose parameters are organ-level, biologically meaningful quantities [19, 20]. Inverting such models, that is, recovering their parameters from observations, has a long history for woody species: Stava et al. fit tree parameters by stochastic search [44], Talton et al. formulated procedural inversion as MCMC inference [45], and Guo et al. inferred L-systems from point clouds [17], with neural variants following [28]. These methods optimize or infer per instance, and none performs amortized, feed-forward inversion of a plant generator from field-degraded input. The closest amortized precedents are outside agriculture: shape-program induction [46], diffusion-based inverse procedural content generation from a single image [54], and cadrille, which maps images, point clouds, or text to executable CAD programs [24, 26]. All operate on rigid, isolated, clean-input objects; none addresses deformable slender organs, inter-plant occlusion, or real-sensor noise, and none studies whether the program parameters should participate in the encoder.

### A.2. Learned parametric recovery of crop plants

Two 2025 systems recover parametric plants from scans. Demeter learns a morphology model (topology, articulation, PCA shape) from real soybean point clouds [8]; Ghrer et al. infer grammar-style parameterized plants from 3D scans [14]. Both take a single, reasonably complete point cloud as input and neither addresses missing modalities or row occlusion. CropCraft recovers crop structure by Bayesian optimization of a procedural model against multi-view NeRF depth [53]; the approach is accurate but per-scene and slow. A staged pipeline annotates terrestrial laser scans with a vision-language model and fits a NURBS procedural descriptor by differentiable refinement [9, 18, 19]; it handles isolated, occlusion-free plants at $\sim$10 minutes each and serves in this paper as both the occlusion-free baseline and the source of real-data procedural ground truth. Tree-D Fusion reconstructs simulation-ready urban trees from street imagery via diffusion priors [29], but its envelope-then-branch pipeline targets woody silhouettes, a different geometry class from thin cereal blades.

### A.3. Point-cloud completion and slender structures

Learned completion runs from PCN [52] through transformer approaches [50, 51] to recent extrapolation–interpolation designs [5], trained and evaluated on benchmarks of compact rigid objects. Neither PCN's eight categories nor ShapeNet-55/34 contains a slender-under-occlusion class, and completion systems assume the missing geometry is self-occluded surface, without clutter from *other* objects inside the crop region. Plant-specific completion is emerging, with amodal segmentation plus completion for occluded plants [27], but it outputs geometry rather than structure or parameters. Work on curve-aware and thin-structure reconstruction [13, 39] addresses the slender regime, in which the standard metric itself fails: Chamfer distance under-weights thin sheets [48], a failure now documented analytically [43] and addressed by density-aware and information-theoretic variants [30, 48]. We further compare geometric metrics with frozen-latent phenotype probes and show that they can rank modalities differently.

### A.4. 3D plant phenotyping

Organ-level 3D phenotyping datasets, including Pheno4D's labeled maize and tomato sequences [42], Crops3D [55], and field-scale maize scans with procedural fits [25], support trait extraction from point clouds [37, 38]. Radiance-field capture is increasingly practical in field conditions [2]. No existing dataset, however, combines real field capture, slender organs, inter-plant row occlusion, and procedural ground truth, and our benchmark is designed to combine all four.

### A.5. Multimodal masked pretraining and distillation

Masked autoencoding [21] extends to multiple modalities: MultiMAE over RGB, depth, and semantics [3]; 4M and 4M-21 to any-to-any generation over tokenized modalities [4, 32]; Point-MAE to geometry [34]; and Embodied-MAE, on which our architecture is based, to RGB, depth, and point clouds for robot manipulation [10]. Missing-modality robustness is an active concern [41], and cross-modal distillation from multi-sensor teachers to single-sensor students is established for discriminative tasks; we distill a *generative* full-modality teacher for structured-parameter regression, and encode continuous parameters with xVal-style number tokens [15]. Our finding that a privileged parameter stream suppresses representation quality connects to modality competition [35] and gradient conflict [7, 49], which offer candidate mechanisms; no prior

work, to our knowledge, documents the effect for generator parameters in multimodal pretraining.

### A.6. Synthetic-data pretraining and sim-to-real

Pretraining without natural data is viable, from abstract fractals [23] to procedural worlds [16, 40], and domain randomization reduces the reality gap [47]. The value of synthetic data is known to be task-dependent [11], and self-consumption is a documented risk [1]. Our setting is the favorable case: the deployment target *is* the generator's parameterization, the generator spans the morphology of the crop, and the method is evaluated on real field data.

## B. Detailed Ablation Studies

All numbers in this appendix are on the final data releases of both species unless marked as first release. Protocol: every arm trains for 197,400 optimizer steps at global batch 32; representation quality is a ridge probe on the frozen CLS token, one fixed view per plant, identical target vectors across arms, with paired bootstrap intervals over plants (a win or a loss needs the 95 % interval to exclude zero on both val and test); generation quality is the validation Chamfer at the matched checkpoint.

### B.1. Modality value-add (E2)

Eight pretraining arms per species differ only in which token streams exist (every stream set that contains the cloud).

### B.2. Token-budget control arms

In the four-stream arm the three sensor streams share the visible budget with the parameter tokens (107.4 visible sensor tokens against 117.0 in the three-stream arm, parameter tokens 58 % visible). The control arms keep the parameter tokens outside the shared budget, restoring the 117.0 sensor tokens, at three parameter visibilities (20 %, 60 %, 0 % = target only). In sorghum no visibility recovers the three-stream arm: height 0.959–0.966 against 0.977, roll 0.702–0.771 against 0.781, and the 60 % and target-only arms lose all nine learnable traits on both splits. In maize the 20 % and target-only arms beat both the shared-budget arm and PC+RGB+D on all eleven traits, while the 60 % arm loses nine. Full table and intervals: Tab. 8 of the main paper.

### B.3. Parameters as a target only

Reconstructing the parameters without ever encoding them (100 % masked as input) still loses to omitting the stream in sorghum: size $-0.011$, roll $-0.07$, mean leaf length $-0.10$, leaf width $-0.15$ (val), intervals clear of zero. Its latent is the most size-dominated of the sorghum arms: the first principal component carries 44 % of the variance and correlates with leaf count at $|r|{=}0.95$, and the participation ratio falls to 4.3 against 6.7 for PC+RGB+D. In maize the same arm is the best four-stream variant and beats PC+RGB+D on every trait, so the effect of the reconstruction target depends on the generator.

### B.4. Data and model scaling at fixed compute

### B.5. Distillation, per source modality

From the headline model of each species, distillation lowers the mean generation loss by 30.2 % in sorghum and 43.5 % in maize (val, epoch-0 warm start against epoch 100). The gain is largest where the warm start is weakest: sorghum Chamfer from the point cloud falls from $5.48$ to $1.39 \times 10^{-3}$ and from the parameters alone from 22.7 to $6.83$; maize Chamfer from the photograph falls from $11.3$ to $0.77$. The parameter error from a point cloud alone halves in sorghum ($0.040 \rightarrow 0.020$) and changes little in maize ($0.020 \rightarrow 0.018$). On the first release, distilling all four targets beat distilling the parameter head alone ($-43.8\,\%$ against $-36.9\,\%$ parameter error), and distillation moved the whole view-difficulty curve down without flattening it (9 of 10 views improved). LR caveat (first release): the parameter specialist ran at $3 \times 10^{-5}$ against $10^{-4}$.

### B.6. View invariance without labels (first release)

Ten views of one plant share one parameter ground truth, so the spread of ten predictions is a label-free view-invariance measure. Distillation roughly halves error *and* view spread on the fields with skill (starting point $-59\,\%$, branching angle $-60\,\%$). Zero-skill fields instead become *more* view-dependent, as the head starts to track image noise; the final generator drops those fields. View stability and accuracy are separate properties: several fields move less than 0.3 of their spread across views while remaining worse than predicting the plant mean.

### B.7. Frozen-encoder baselines and camera-pose forensics

On sorghum the three-stream arm beats every published frozen encoder (EmbodiedMAE-B [10], Point-MAE [34], DINOv2 [33], MultiMAE [3]) on every trait, and EmbodiedMAE-B, the strongest, on eight of nine paired over plants (size $+0.004$ to $+0.007$, roll $+0.04$, leaf length $+0.08$, leaf width $+0.09$); it also beats a network of our architecture trained from scratch on the probe targets on seven of nine traits. On maize the four-stream arm beats EmbodiedMAE-B on 10 of 11 traits and the best pose-free baseline per trait on 9 (val, point estimates), but the maize network trained from scratch is ahead of every frozen 600-epoch arm on all eleven; the maize headline model closes most of that gap. On the first maize release, supplying Point-MAE with each view's full camera pose gave it the generator's fixed plant azimuth, a canonicalization that no field rig provides; against Point-MAE given gravity alone (the information a calibrated rig provides), our pose-free

| Arm | Sorghum Chamfer$\times 10^{-3}$ | Sorghum height $R^2$ | Maize Chamfer$\times 10^{-3}$ | Maize height $R^2$ |
|---|---|---|---|---|
| PC | 5.00 | 0.895 | 2.63 | 0.171 |
| PC + depth | 2.45 | 0.976 | 2.55 | 0.453 |
| PC + RGB | 2.57 | 0.979 | 1.61 | 0.620 |
| PC + RGB + depth | 1.85 | 0.977 | 1.21 | 0.633 |
| PC + params | 4.84 | 0.860 | 2.49 | 0.256 |
| PC + depth + params | 4.08 | 0.907 | 2.54 | 0.382 |
| PC + RGB + params | 3.00 | 0.962 | 1.21 | 0.509 |
| all four | 2.53 | 0.962 | 0.99 | 0.635 |

Table 11. E2, val. In sorghum the parameter stream raises Chamfer in every set except PC alone and lowers the probe; in maize it lowers Chamfer but its effect on the probe depends on the set (added to PC: 7 wins, 0 losses of 11; to PC+RGB: 3 wins, 5 losses; to PC+RGB+D: a draw). Full probe columns are in the main E2 table.

| | Sorghum Chamfer$\times 10^{-3}$ | Sorghum height $R^2$ | Maize Chamfer$\times 10^{-3}$ | Maize height $R^2$ |
|---|---|---|---|---|
| 1,000 plants | 3.13 | 0.952 | 3.54 | 0.393 |
| 3,000 plants | 2.86 | 0.932 | 1.46 | 0.524 |
| 10,000 plants | 2.48 | 0.962 | 0.87 | 0.427 |
| 10,500 (all) | 2.53 | 0.962 | 0.99 | 0.635 |
| small (25 M) | 2.86 | 0.964 | 1.22 | 0.563 |
| base (114 M) | 2.53 | 0.962 | 0.99 | 0.635 |
| large (332 M) | 2.68 | 0.948 | 1.06 | 0.580 |

Table 12. Four-stream arm, 197,400 steps per row, val. Paired over plants in sorghum, the 10,000-plant arm ties the full-data arm (one win, no loss), the 3,000-plant arm loses six traits and the 1,000-plant arm four. In maize every reduced-data arm loses 10 of 11 traits to full data and the probe is not monotone in data (the 3,000-plant arm beats the 10,000-plant arm on height, leaf length and leaf count). Base is the best size in both species; in maize small and large each lose 10 of 11 traits to base.

model won 5 of 9 decided targets. Tassel droop is the weakest trait for every encoder of this design.

### B.8. Latent structure (E9)

The latents are continua, as expected for a single-generator population: best silhouette 0.15 to 0.24 at $k$=2, adjusted Rand index $\approx$ 0 against shape deciles, t-SNE trustworthiness 0.95 to 0.99. The participation ratio is 4.4 to 6.3 (of 768) in sorghum and 6.7 to 11.3 in maize, consistent with maize's more independent generative factors, and the dominant principal component of the sorghum latents is the plant's size. Linear decodability is distinct from local embedding structure: the probe reads roll at $R^2$ 0.78 (PC+RGB+D, val) from a latent in which roll is barely organized locally. We therefore use probes, not t-SNE smoothness, for representation claims on continuous generators.

## C. Dataset Generation

FIELDPROC holds four synthetic sets: isolated single-plant captures of maize and sorghum, and *row views* in which the same plants stand in a crop row between two neighbors and are photographed by a ground rover. All images are $1024 \times 1024$ with a $40^\circ$ vertical field of view ($f_x = f_y = 512/\tan 20^\circ = 1406.7$ px, principal point at the center), in the OpenGL convention ($y$ up, the camera looking down $-z$).

### C.1. Procedural plants

**Maize.** Plant $i$ of 15,000 is generated with seed $i$ by a procedural maize generator whose traits are drawn from Gaussians (height $0.832 \pm 0.104$ m, leaf count $12.35 \pm 2.57$, base leaf length $0.448 \pm 0.057$ m, base leaf width $0.066 \pm 0.010$ m, base leaf angle $43.98^\circ \pm 10.13^\circ$), coupled through four latent traits (vigor, slenderness, posture, texture) with correlation strength 0.735, with bottom, middle and top canopy curves and per-leaf jitter (scale 0.687). The plants are 0.62–1.44 m tall (median 1.01 m). The generator record is reduced to the plant height, the leaf count and, per leaf, its distance along the stem, length, width, azimuth, angle and droopiness; per-leaf curvature is added by rebuilding the midrib B-spline (target elevation angle$+$droopiness$\cdot t^2$) and integrating $\theta = \int |\kappa|\, ds$ over the blade, $s \in [0.22, 1]$, with the blade length $S$ from 10-node Gauss–Legendre quadrature.

**Sorghum.** Plant $i$ of 15,000 is generated with seed $i$ by the EvoEngine sorghum generator. It draws the plant heading uniformly in $[0°, 360°)$, the stem tilt ($0 \pm 1.5°$), the leaf count ($20 \pm 2$, clamped to $[2, 128]$) and the internode length ($0.04 \pm 0.03$ m), and sets the stem length to internode $\times\, n_{\text{leaves}}/(1 - s_0)$, with $s_0$ the first leaf's starting point. Leaf $k$ at $t = k/(n-1)$ takes its starting point, length, width, roll ($\text{roll}_k = (k \bmod 2) \cdot 180° + \text{roll}(t)$), branching angle, bending, curling and waviness from curves over $t$. No panicle is generated. The plants have 4–35 leaves (mean 19.5) on stems of 0.19–1.74 m.

## C.2. Splits

Both species are split by plant into 10,500 / 2,250 / 2,250 plants. *Maize:* 19 scalar features are robustly standardized, $z = (x - \text{median})/(1.4826\,\text{MAD})$, each plant is scored by $\sqrt{z^\top (\Sigma + 10^{-6} I)^+ z}$, and plants are dealt from most to least extreme alternately to test and validation until both hold 2,250 (seed 0). Train scores span 1.56–4.60, validation 4.60–23.3, test 4.60–32.2. *Sorghum:* the extremeness reproduces, to the stored precision, as the Mahalanobis distance from the mean (sample covariance) over leaf count, stem length, mean and maximum leaf length, and mean and standard deviation of leaf roll. The hold-out probability rises with a plant's extremeness rank, from 0.3 % in the least extreme 5 % to 70 % in the most extreme 5 % (median rank 0.75 on validation and test, 0.37 on train). This rule is inferred from the released assignment; the script that wrote it was not found.

## C.3. Isolated-plant captures

Both species are rendered off-screen with `pyrender`. The camera looks at the plant's bounding-box center $\mathbf{c}$ with world up $(0, 1, 0)$; lighting is ambient 0.15 plus three white directional lights aimed at $\mathbf{c}$ from $(1, 1, 1)$, $(-1, 0.5, -0.5)$ and $(0, 1, 0)$ (intensities 4.0, 2.0, 1.5), on a gray (0.5) background. Depth $z$ with near and far planes $n, f$ is stored as $v = \lfloor \text{clip}((z - n)/(f - n), 0, 1)(2^{32} - 1) \rfloor$ across the four 8-bit channels of a PNG, most significant byte first; $v = 0$ is background. A pixel $(u, v)$ with depth $z > 0$ back-projects to

$$\mathbf{p}_{\text{cam}} = \left( \tfrac{u - c_x}{f_x} z,\ -\tfrac{v - c_y}{f_y} z,\ -z \right), \tag{8}$$

and clouds are drawn uniformly without replacement from these points.

*Maize:* ten view directions per plant are drawn uniformly on the sphere (seeded per plant), at distance $1.05\, r / \sin 20°$ with $r$ half the bounding-box diagonal; a fourth light (1.5) is placed at the camera, and textured leaf, stem and tassel materials are double-sided. Each view stores RGB, an RGBA image with the background cleared, depth, the camera pose, the trimmed parameter file, and an 8,192-point cloud back-projected from that view's depth (Eq. 8), so the cloud contains only the surface visible to the camera. *Sorghum:* the ten directions form a Fibonacci sphere, $y_i = 1 - 2(i + \frac{1}{2})/10$, $\phi_i = i(\sqrt{5} - 1)\pi + \epsilon_i$ with a small fixed jitter, identical for every plant (five above and five below the horizon), at 2.5 m; one leaf texture covers the plant and there is no camera light. Each view stores RGB, depth, the camera pose, the parameter file, and the complete, occlusion-free labeled plant cloud moved into the camera frame.

## C.4. Row views

**Composition.** Within a split with plant list $L$, row $i$ has main plant $L_i$, right neighbor $L_{i+1}$ and left neighbor $L_{i+2}$ (indices modulo $|L|$), so every plant is a main plant once. The world frame is the main plant's (stem base at the origin, row along $x$, rover on the $+z$ side), with the neighbors' stem bases at $(\pm d, 0, 0)$, $d = 0.15$ m. Each row is rendered at ten orientations: the main plant is turned about its stem base by $\theta \in \{0°, 10°, \ldots, 90°\}$, each neighbor by $\theta + \delta$, $\delta \sim U(-15°, 15°)$ (seeded per split and row). Sorghum stems lean, so each sorghum neighbor is also rotated about its stem base by the smallest rotation that makes its stem axis (the principal axis of its stem vertices) parallel to the main plant's; the main plant is never tilted, and maize is not tilted.

**Contact solver.** The three plants are made free of intersections while staying as close as possible to their grown shapes; they may touch but not cross. Each blade is a chain of 8 cm segments from collar to tip whose joints bend along circular arcs, up/down and sideways, so the blade stays smooth, keeps its length and stays attached. Stems and sheaths (everything within 2.5 cm of the stem axis) and the maize tassel are rigid. A leaf's joint bends sum to a total-turn limit, and a leaf may not bend into its own stem or cross its own plant's leaves more than it does as grown. Where a leaf would cross another plant, it bends at the joint before the contact by the smallest clearing angle, smallest first, so it comes to rest on or under what it meets. Neighbors bend first; a main-plant leaf bends only where a neighbor cannot move (a neighbor's stem or sheath, or a neighbor leaf with no clear shape). A sorghum neighbor leaf that cannot bend clear may also roll about its stem by up to $\pm 90°$. Candidates are screened on 1 cm occupancy grids and every remaining contact is decided by exact triangle–triangle tests; a found layout is then settled, each bent leaf easing back toward its grown shape to the least bend that still clears. Unsolved layouts are retried under looser limits (Tab. 13), then at spacings raised in 5 cm steps up to 1.5 m and refined in 1 cm steps. An independent NumPy triangle–triangle test on the rendered positions found no crossing plant pairs on the rows checked.

| Stage | collar side | blade side | up/down | total turn |
|---|---|---|---|---|
| 0 | 30° | 10° | 45° | 90° |
| 1 | 60° | 10° | 60° | 120° |
| 2 | 90° | 30° | 70° | 180° |

Table 13. Staged bend limits of the contact solver (per joint, except the total turn per leaf).

**Rover camera and outputs.** One shot is taken per orientation from $(0, h, \mathrm{dist})$, looking horizontally at $(0, h, 0)$ with $h$ the height of the main plant's bounding-box center; $\mathrm{dist} = 1.0\,\mathrm{m}$ for maize and 2.5 m for sorghum, near 0.05 m, far $\mathrm{dist} + 2\,\mathrm{m}$, with the lens, lights and materials of the isolated captures. Each shot is rendered twice from the same camera: the input with all three plants (RGBA, depth, and a 24,576-point cloud of every visible plant) and the ground truth with the main plant alone in its bent shape (RGBA, depth, 8,192 points), with the main plant's parameter record. The maize input clouds were first produced at 8,192 points and rebuilt at 24,576 from the stored depth; the old clouds lie within $8.4 \times 10^{-8}$ m of the rebuilt ones.

| Split | shots | at $d$=0.15 m | stage 0 / 1 / 2 | max $d$ |
|---|---|---|---|---|
| train | 105,000 | 76.4 % | 81,679 / 16,138 / 7,183 | 0.33 m |
| val | 22,500 | 78.3 % | 17,780 / 3,321 / 1,399 | 0.28 m |
| test | 22,500 | 79.0 % | 17,824 / 3,284 / 1,392 | 0.30 m |

Table 14. Maize row statistics. The 99th percentile of the spacing is 0.23 m in every split, and no orientation failed. On 10 sample sorghum rows (100 orientations), 85 were solved at 0.15 m and the rest at 0.16–0.21 m.

**Caveats.** The row ground truth is the main plant as bent in that row, while its parameter record describes the grown plant; main-plant leaves are bent only where unavoidable. At 2.5 m the lens covers about 1.8 m of height, so the top or base of tall sorghum plants can leave the frame (in the 10 sample sorghum rows the main plant touches the top edge in 58 of 100 shots). The two isolated sets differ in view sampling, camera distance, materials, lighting and per-view cloud. Sorghum plant 0 uses seed 0, for which the generator does not reseed. Check that plant 0's parameters match its mesh.